Laboratory Manual

Rock Excavation Practical

M.Tech (TUST and OCM) : MNC 516

B.Tech. (ME) : MNC 204

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Disclaimer

This laboratory manual designed and compiled has contents from several sources including indigenous design and developments, lecture notes, fields cases/data, various related company websites available in public domain, journals, texts etc. Most of the references are quoted in the concerned experiments. Any omission is inadvertent only. This course lecture material is meant for internal & private circulation of the concerned practical course participants only. The practical materials, figures and photographs should not be copied or used for any purpose without due referencing or acknowledging properly. Mentioning of a few company names may not be construed as a recommendation to that product. It is only for providing typical idea and features to only educate the students. Any suggestions for improvement in the manual are welcome.

M.Tech.(Opencast Mining)

M.Tech.(Tunnelling and Underground Space Technology)

Course Type	Course Code	Name of Course	L	T	P	Credit
DC	MNC516	Rock Excavation Practical	0	0	3	3

Course Objective

The objective of this practical is to enable the learner to understand the determination of different physico-mechanical properties of the intact rock, soil and joint as per the suggested methods (ISRM, ASTM, BIS etc.) and also provide an over view of their application in geo-engineering design.

Learning Outcomes

Upon successful completion of this course, students will:

- know determination of various mechanical properties (dynamic)
- correlate their application in rock excavation system design

List of practicals

1. Sonic velocity and dynamic elastic constants
2. Field seismic velocity and analysis
3. Blast vibration measurement and analysis
4. Measurement of VOD of explosive and analysis
5. Fragmentation/chip size analysis
6. Burden velocity measurement and analysis using high speed camera
7. Fracture toughness
8. Cerchar hardness and Cerchar abrasivity indices
9. Drilling rate index
10. Punch Penetration Index
11. Linear cutting rig experiments
12. Bit wear index, Abrasion value steel and cutter life index

Text Books:

1. ISRM and other (NTNU, CSM, ISEE) Suggested Methods
2. Jimeno, C.L. et al., Drilling and Blasting of Rocks, A.A.Balkema, 1995
3. Tunnel boring machines: Trends in design and construction of mechanized tunnelling, Ed. H.Wagner & A.Schulter, AAB, 1996
4. ISRM Suggested Testing Methods : E.T.Brown
5. Web Page of ISRM: https://www.isrm.net/gca/?id=177
6. Design Analysis in Rock Mechanics, W.G.Pariseau, Taylor & Francis, 2012

(This is the order followed in detailing the practicals in this lab manual)

B.Tech.(Mining Engineering)

Course Type	Course Code	Name of Course	L	T	P	Credit
DP	**MNC 204**	**Rock Excavation Lab.**	0	0	2	2

Sl. No.	Major Topics	No. of Practicals	Learning Outcome
1	Sonic velocity in rocks	1	Dynamic elastic constants
2	Field seismic velocity	1	Shallow seismic refraction for subsurface characterization
3	Cerchar hardness index	1	Resistance to indentation, drilling rate estimation
4	Punch penetration index	1	Rock toughness
5	Cerchar abrasivity index	1	Wear of cutting tools
6	Drilling rate index	1	Rock drillability and drill selection
7	Blast vibration measurement	1	Seismic impacts on structures due to blast induced ground and air vibrations
8	VOD measurement	1	Explosive selection and performance estimation
9	Burden velocity measurement	1	Dynamic analysis of moving burden for delay selection and fragmentation/vibration control
10	Blast design and fragmentation analysis software	1	Blast design simulation and fragmentation assessment through image analysis
11	Linear cutting rig tests	1	Specific energy estimation in rock cutting

Text Book:

1. ISRM and other (NTNU, CSM, ISEE) Suggested Methods
2. Jimeno, C.L. et al., Drilling and Blasting of Rocks, A.A.Balkema, 1995
3. Tunnel boring machines: Trends in design and construction of mechanized tunnelling, Ed. H.Wagner & A.Schulter, AAB, 1996

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Do's and Don'ts

The following instructions may be followed while conducting practicals in the above-mentioned laboratory:

1. Read the experimental manuals/procedures supplied to you in advance, carefully and completely and make your own observations on the type, procedure and care that needs to be exercised in each practical.

2. Listen to the instructions of the instructor, technical personnel and associated scholar and follow meticulously. Any doubts with regards to usage please do not hesitate to contact any one of the above.

3. Unless you are asked do not touch any equipment, samples or any part of the equipment.

4. Carryout the measurements and experiments under the supervision of the designated instructor. Do not operate or insist to operate equipment without proper permissions.

5. Do not leave any of your belongings in the laboratory and check before you leave.

6. Take a good care of yourself, the equipment and samples and leave them in good shape.

7. Any uncommon noise, smoke, vibration and leakage while operating the machine or any other nearby equipment if noticed please bring it to the notice of the concerned laboratory technical person and teacher concerned. Never venture into solve it unless asked for.

8. Any impeding danger foreseen please evacuate the place and inform the concerned.

9. Wear cotton clothes with loose fit and shoes before you enter the lab. Goggles as per the need shall be provided to you to wear. Wear them before you start your work.

P1- SONIC VELOCITY AND DYNAMIC ELASTIC CONSTANTS

Aim: Determination of sonic velocity and dynamic elastic constants of rocks

INTRODUCTION

This test is used to determine the velocity of propagation of elastic compression (Vp) and shear waves (Vs) in rock samples in the laboratory. Primary waves (P-waves) are compressional waves that are longitudinal in nature and vary with the lithology, porosity, and bulk density of the material; state of stress, such as lithostatic pressure; and fabric or degree of fracturing. Together with density measurements, sonic velocity is used to calculate acoustic impedance, or reflection coefficients, which can be used to estimate the depth of reflectors observed in seismic profiles and to construct synthetic seismic profiles. The ultrasonic (dynamic) elastic constants of a rock can be determined from the compression and shear wave velocities and the rock density for isotropic or slightly anisotropic rock.

TEST PROCEDURE

Connect the Sonic viewer 170 to the transducers as shown in Fig. 1 (one is work as a transmitter and other is receiver)

Fig. 1: Sonic viewer with transducer

Now, the step relates to ZERO Adjustment with the transducers. For extending the velocity measurement with the sample (the test specimen), setting in advance is required with ZERO point with the transducer response. Firstly, pile up the transducers as illustrated in Fig.2. Then, after setting Gain, Filter, Range etc, press the key READY to start with measurement. The menu like the one indicated in Fig. 3 is repeatedly and continuously displayed. Pressing the key ESC works to stop the measurement.

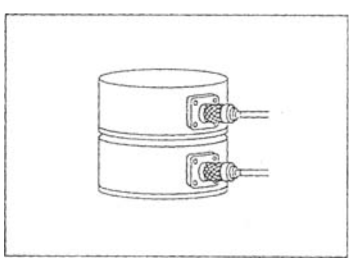

Fig.2: (Zero adjustment)

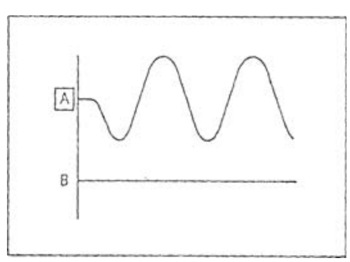

Fig.3: (Zero adjustment)

Then shift Cursor key to the position of the First Break. At this position of the First Break represents the response time of the transducer, fix this time as ZERO point. Place the sample as measurement object (test specimen) between the transducers, as illustrated in Fig. 4. In case of the measurement through the use of P-wave transducers, apply silicon grease to the contacting faces between the transducers and the sample.

Fig. 4: Setting the Sample Core

Then, after setting Gain, Filter, Range etc, press the key READY to start with measurement. When the waveform data like the one shown in fig.5 has been obtained during the measurement, press the key ESC to stop the measurement or alternatively, re-press the key READY.

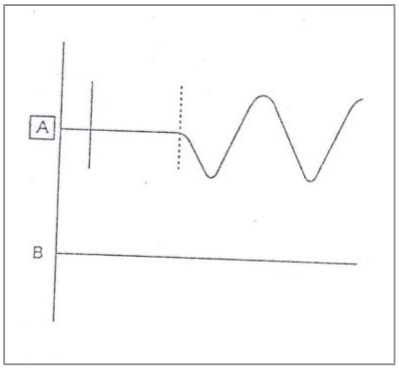

Fig.5: Measured Waveform

Shift cursor key for adjusting the position to the one of the First Break for sample measurement. The arrival time of the First Break with sample are displayed in the columns of TA, TB in the upper part of CRT. Press the key LENGTH on function switch to display menu fig.6.Then after putting the value of length of test specimen, press the key VP of Data switch to get P-wave velocity.

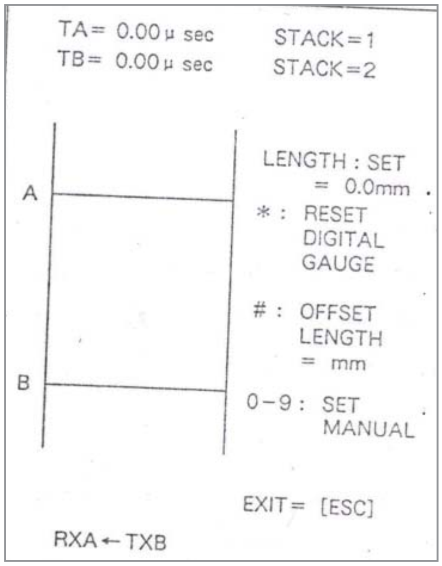

Fig.5: Menu

CALCULATION

The velocities are calculated from the travel time (tp and ts) and the distance between transmitter and receiver (d), by using the following equations:

Velocity compressional wave (Vp) = d/tp (m/s)

Velocity shear wave (Vs) = d/ts (m/s)

The following ultrasonic (dynamic) elastic constants can be calculated if the density of the material is determined and if the sample is isotropic or very slightly anisotropic:

$$\text{Modulus of elasticity} \quad E = \frac{D * Vs^2 * (3Vs^2 - 4Vs^2)}{Vp^2 - Vs^2} (\text{in Mpa})$$

$$\text{Modulus of rigidity} \quad G = D \, x \, Vs^2 \, (\text{in Mpa})$$

$$P'\text{oissons ratio} \quad \nu = \frac{Vp^2 - 2Vs^2}{2(Vp^2 - Vs^2)}$$

$$L'\text{ames constant} \quad \lambda = D\,(Vp^2 - 2Vs^2)\,(\text{inMpa})$$

$$\text{Bulk modulus} \quad K = \frac{D\,(3Vp^2 - 4Vs^2)}{3} (\text{inMpa})$$

Where, D = Rock material Density (Kg/m^3)

The observation table for recording the results of ultrasonic velocity is given below:

Rock type	Density	Time (Micro sec)	V_P (km/sec)	Time (Micro sec)	V_S (km/sec)	E (GPa)	Poisson's ratio
Marble	2.72	10.4	5.77	39.0	1.54	18.87	0.46
	2.67	10.0	5.89	40.0	1.47	16.89	0.47
Quartzite	2.71	25.6	4.01	51.0	2.01	29.19	0.33
	2.69	22.4	4.00	55.0	1.63	19.99	0.40
Marble	2.64	11.6	2.54	37.0	0.80	4.89	0.44
	2.58	14.0	2.32	34.0	0.95	6.53	0.40
	2.75	42.8	2.43	78.0	1.33	12.53	0.29
Charnockite	2.62	8.0	5.58	26.0	1.72	22.45	0.45
	2.65	7.6	5.97	32.0	1.42	15.70	0.47
Shale	2.51	18.4	2.48	34.0	1.34	11.67	0.29
	2.44	20.4	2.23	44.0	1.03	7.07	0.36
	2.36	6.8	3.53	20.0	1.20	9.77	0.43
	2.64	7.2	4.16	20.0	1.50	16.90	0.43

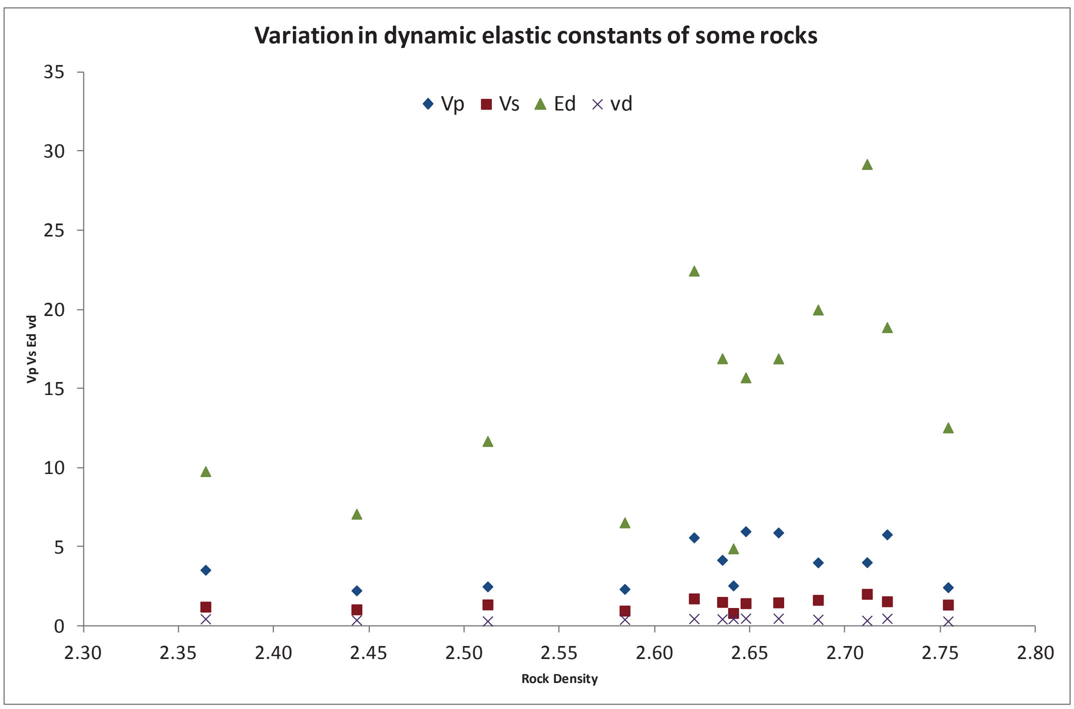

Fig.6: Typical variation in dynamic properties is shown against density of rocks

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P2 - FIELD SEISMIC VELOCITY AND ANALYSIS

Aim: To determine the insitu P wave velocity of rock layers for various rock excavation applications

Introduction: seismic techniques are non-destructive and easy to apply; they are increasingly being used in geotechnical engineering. Seismic tomography is a process for obtaining images or maps of seismic velocity or other seismic properties of a rock mass. Seismic tomography offers the opportunity to investigate indirectly the in situ, bulk geomechanical properties of the rockmass away from surface or borehole exposures. Seismic velocity images reveal variation in elastic moduli which, in turn, are related to several properties required in the geological or geomechanical assessment of a rock mass. Seismic velocity is sensitive to lithology, discontinuities, and in situ stress. A close correlation between blastability indices, excavatability indices and velocity of longitudinal wave have been shown by many researchers.

Seismic techniques

There are two types of seismic techniques, **Reflection technique and Refraction technique.**

The principle deference between the geometry of the refraction and that of the reflection methods is in the interaction that takes place between the seismic waves and the lithological boundaries they encounter in the course of their propagation.

Reflection technique: Reflection technique is used to **detect** the structural features or weak planes.

Refraction technique: Refraction technique is used to find out the **magnitude** of the structural features or weak planes. Refractions are relatively low frequency because of the greater distance travel and usually include more cycles than reflections. Refraction paths cross boundaries between materials having different velocities in such a way that energy travels from source to the receiver in the shortest possible time. Most refraction surveys involve the use of waves having trajectories along the tops of the layer with speeds that are appreciably greater than those of any overlying formations. The shot and geophones must be farther apart for refraction than they would be for reflection.

Principle:

Seismic waves when passes through various mediums having different velocities, get refracted. The law of refraction states "if the ray incident on the interface between the two media makes an angle i_1 with the normal to the interface, the refracted ray in the adjoining medium makes an angle i_2 with the normal"

$$\text{Sin } i_1/\text{Sin } i_2 = V_2/V_1$$

Where,

V_1 and V_2 are the seismic velocities in the two media respectively

If V_2 is greater than V_1, we have Sin i_2>Sin i_1 and therefore $i_2 > i_1$.thus, the refracted ray makes a larger angle with the normal i.e. smaller angle with the interface than the incident ray.

If the incident ray makes a particular angle i_c such that i_2=90 (sin i_2=1), then

$$\text{Sin } i_c = V_1/V_2$$

In this case, the refracted ray travels along the interface and the angle of incidence is called the critical angle. The surface wave travels slower than either body waves and is generally complex. Any wave meeting an interface between media of differing velocities is partly reflected and partly refracted. In Figure 1, the full line is the critically refracted ray path. The broken line shows successive position of the "head wave", which is the surface expression of the refracted P or S wave front. The surface wave, although slower, reaches the nearest geophone first if it is within a critical distance from the shot point. The P wave reaches the geophone first if it is beyond this distance. By knowing a distance from the shot point to the geophones and the travel time, it is possible to calculate the seismic velocity of the near surface medium and of the next lower layer of material. Typical P wave velocities in different materials is presented in Table 1.

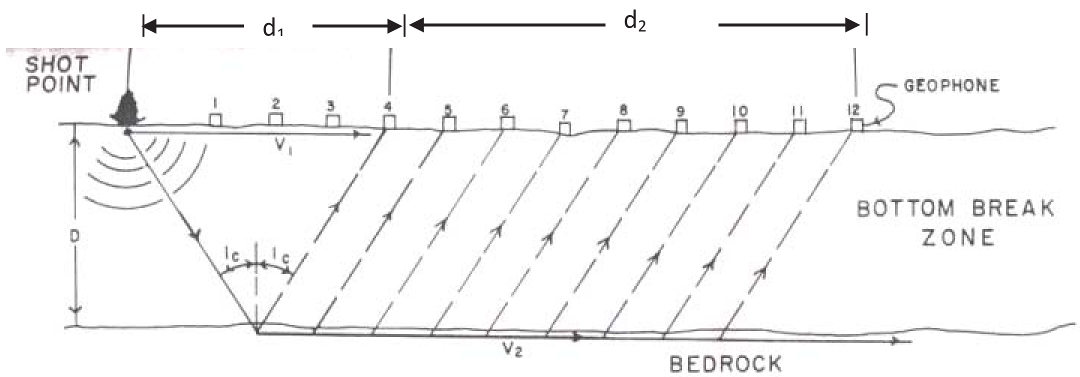

Fig.1 – Seismic refraction process

Table 1- P – wave velocity of different rock masses (P.Pal Roy, 2005)

Name of rockmass	Rock type	P – wave velocity
Dunite	Compact	7000
Diabase		6500
Gabbro		5600
Dolomite		5500
Granite		5000
Limestone	Less Compact	4000
Slate & Shale		4000
Sandstone		3000
Alluvium	Unconsolidated	1000
Loam		1000
Sand		1000
Loess		500

Seismic imaging using Geode (8 Channel)

Specifications of Geode

- ***Geode is a highly portable stand alone distributed seismic module.***
- Weight of Geode= 6 to 8 pounds = 2.72 to 3.63 kg.
- Geode has 20 kHz bandwidth
- 12 –volt battery is required with the setup.
- Geode uses about 0.6 W per channel while operating but can be put into power save mode between shots which reduces power consumption to about 0.1 W per channel

Accessories with Geode Seismic System

- SGOs (Single Geode Operating Software) operating software.
- Power cable with alligator clips.
- Ethernet interface card for inserting into the laptop.
- Geode Digital Interface cable.
- Hammer / Trigger switch (typically attached to the energy source like a hammer).

Other Recommended Accessories for a seismic survey

Laptop computer to control the Geode.

- Geophones (typically 3 or more depending on configuration)
- Spread cable (with connections for attaching the geophones).
- Trigger cable (to communicate a trigger start to the geode).
- Seismic energy source: hammer, drop weight, explosive
- Tape measures and survey equipment are necessary to ensure accurate positioning of the geophones.

Connections and working of Geode (also some precautions)

Connections of the Geode setup for a seismic survey are shown in Figure 2.

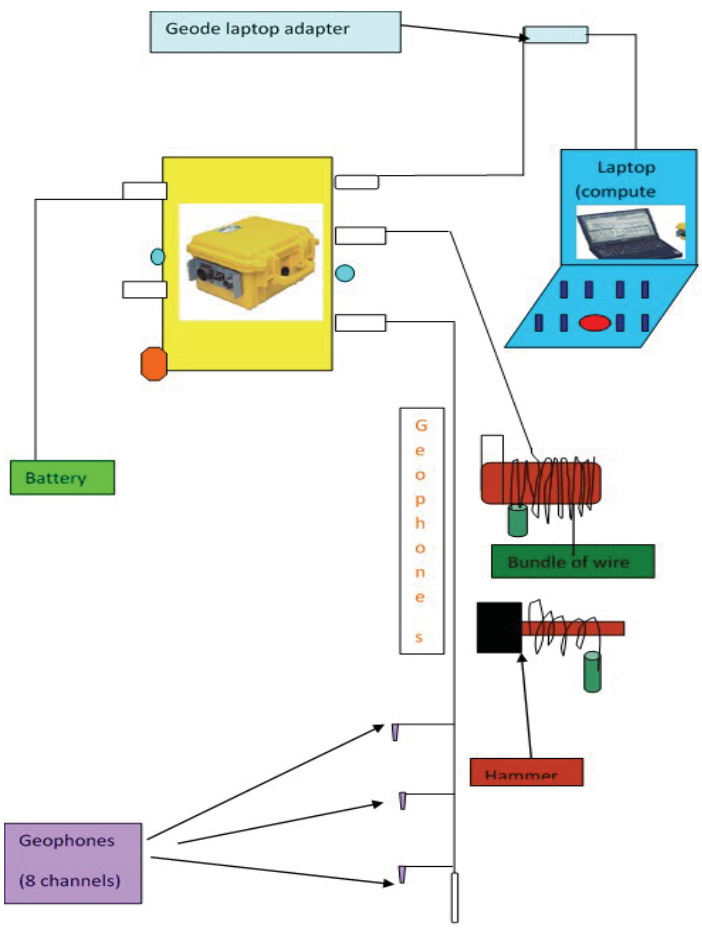

Fig.2 – Schematic layout of Geophone array and Geode connection

Plant the geophones firmly (by pressing the spike into the ground) by each connector (called a takeout) on the cable. If the ground is very soft so that when planting the geophones there remains airgap between ground and geophones then plaster of Paris can be used. Connect the geophones to the spread cable. The geophone connectors will have a method to encourage proper polarity connections (such as wide and narrow colour- coded clips).

Starting the Geode

First turn on the power switch near the power connector on the geode. The adjacent blue LED will start flashing every 3 seconds, indicating that the Geode is in the standby mode. The blue light should be clearly visible, even in bright Sun. Next start the laptop. After the laptop has been turned on and network card is active, the bright blue LED near the digital interface output cable attached to Geode starts to flash every 3 seconds if the interface is working properly. Start the operating software. The LED's on the Geode will briefly flash very quickly, indicating that the software is being downloaded and the circuitry is being initialized.

Geode operating software main screen.

When the seismic operating software starts, a screen comes on laptop as shown in Fig.3.

- Noise Monitor window that continually displays the output from the geophones,
- Shot window that will be initially blank that will display the results of the data acquisitions
- Log window that displays a record of your survey and the activities that you have undertaken. The log file is handy for reviewing the events that happened during course of a survey, particularly if there is any confusion later concerning what might have taken place.

Menu Structure and Getting Around

A menu bar at the top of the screen is ordered from left to right to follow a typical survey setup procedure as shown in the Fig.3.

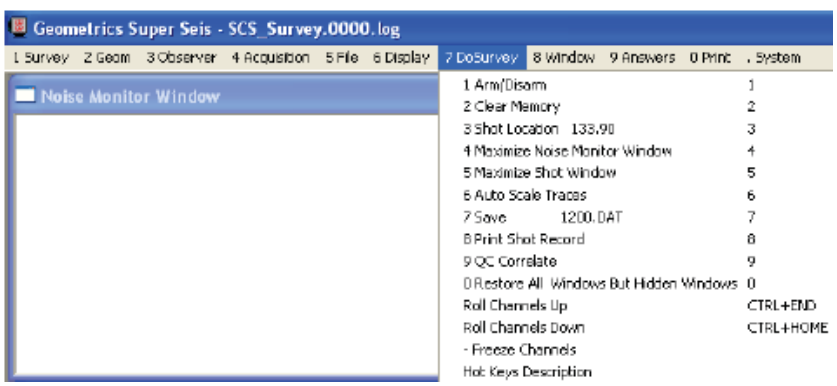

Fig.3: Showing menu structure of the Geode and software

Survey sets up the survey name

Geom sets up the type of survey to be done, the position of the shots and geophones and the pattern of movement

Observer stores accounting information about the survey

Acquisition configures the seismic recording parameters like record length and sample rate

File defines storage parameters

Display determines how the data is displayed

Do Survey contains the majority of operations for undertaking a survey. Most operators will spend most of their time in this window after initial set up

Answers processing and interpretation options

Print configures hard copy output

System configures the trigger, allow setting of the date, attaching of other Geodes etc.

Beginning a Survey

Position the cursor over **Survey** in the top left corner of the screen. It will be asked to start a **New Survey.** Enter a *Survey Name* and *Line Number that are pertinent to the job*. It can be seen that these entries are reflected in the Survey Log window, to help keep track of the progress of the survey. Thus, we will be able to review the log file at a late date if we need to recall anything.

Setting system parameters:

Geometry

The location of the shot and the geophones must be recorded each time data is acquired with the Geode. First set the distance between geophones and the shot location. Select the **GEOM** menu from the menu bar. Next set the **Group Interval.** The Group Interval or geophone spacing refers to the spacing between geophones. For the field study the spacing between 2 geophones has been kept at 6 meters all throughout. 8 Geophones were always used for the field study. Next, make sure **Shot Increment** and **Geophone Increment** are set to zero.

Acquisition: In seismic data acquisition, electrical signals from the geophones are amplified, digitized, and stored in the seismograph's memory. First, select **Sample Interval/Record Length:** Set a *Sample Interval* of 0.250 ms and a *Record Length* of 0.200 seconds. Leave the *Delay* at zero and select **OK**. Next, set how channels are used in the **Specify Channels** menu. It can be seen that this is the same menu that was set up in the **GEOM** section, but this time make sure that 8 channels in the **Use** section are set to **DATA.** This makes 8 channels active. Finally, the last item to set up in this section is **Preamp Gains.** For the purposes of the field study **All 36 dB was** selected**.**

File : Select **File** then **Storage Parameters** from the main menu bar . Set the **Next File Number** to say 437(for example). This will be the name of the first file that is saved. It will increment automatically with each subsequent save. Disable **AutoSave** and ensure that **Save to Disk** is checked so data can be written and select a **Path** to drive D. Note: The file number should be a numerical integer, not alphanumerical. The read disk function in the file menu allows to read previously stored data into memory for display on the screen. Simply choose the path and file name of the file that need to be analysed and press OPEN.

Do Survey (Fig.4)

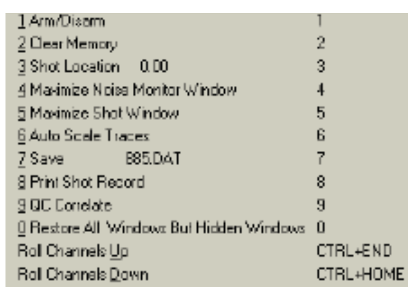

Fig.4: Window of do survey menu

Answers: The **Answers** menu provides refraction interpretation tools. Also, third-party software can be run from here. Remember, in order to use the utilities in this menu, enter the geometry accurately in the **geometry** menu.In this menu the sub menus are Pick breaks, Solve Refraction Using SIPQC, Launch optimum @ survey, Launch Oyo first break picker and Launch Oyo Refraction Analyser. The first step in analyzing refraction data is to identify the first arrivals of seismic energy on each trace. These first arrivals are identified as the first position where the trace deflects from a straight line. There is an automatic first break picker in the software that will help to identify the first break position. However, it will only work well when the picks (first arrivals) are very distinct otherwise first breaks can be picked manually. The trace for the geophone nearest the impact point should have clear, early arrivals. First arrivals on traces from geophones further away should occur progressively later in time. After all of the first arrivals of energy have been selected, the picks can be saved on disk by pressing **Enter.** The pick file will be saved with the same name as your data file, but with the extension '.bpk.'. These files can be retrieved later for interpretation.

Launch optimum @ survey: For getting the field insitu p-wave velocity interpretations by Launch optimum @ survey click on its menu. Then saved files of" bpk" be chosen (6 files –3 shots in normal spread and 3 in reverse spread can be selected). It is the easiest and fastest of the interpretation tools.

The PickWin Module: The main purpose of PickWin is to help in identifying the first breaks, pick them, and save them for input to the analysis program, PlotRefa.

The PlotRefa Module: PlotRefa is the interpretation module of SeisImager. It takes the output of PickWin as input, and through the application of one of the three available interpretation techniques, provides a velocity cross section. It includes many useful tools for facilitating data interpretation.

ISRM suggested important guidelines for conducting seismic refraction survey (Takahashi T.2004. "ISRM suggested methods for land geophysics in rock engineering". International Journal of Rock Mechanics & Mining Sciences 41(2004) pp:885-914.

Arrangement of the survey lines: Because the seismic refraction method derives two-dimensional depth profiles, the survey lines should be arranged perpendicular to the strike of the target geological structures and boundaries. Positioning of the survey line: If there is more than one spread of Geophones in a line, the ends of each spread should overlap so that continuity in the travel time data can be preserved. Preparation for the measurement: Prior to giving the shot command to strike the hammer, the operator of the recording equipment needs to monitor the background noise level and to ensure that noise from wind, from road traffic, from drills and shovels, from aircraft, etc, is at a minimum.

Measurement of insitu P-wave velocity and seismic imaging

A scanline was spread on the bench. In the main spread cable Geophones were connected via take outs. The scan line was approximately in the middle of the bench as can be seen in Fig.5. Some photographs showing various rock types in the exposed bench are given below. From the photograph firsthand understanding of the conditions of the bench can be obtained.

Fig.5- Data acquisition and analysis for insitu seismic velocity

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P3-BLAST VIBRATION MEASUREMENT AND ANALYSIS

Aim: Measurement of blast-induced ground and structural vibrations and analysis

Ground Vibrations

When a explosive charge detonates in the blasthole intense strain waves are transmitted to the surrounding rock. The energy carried by these strain waves known as strain energy, fragments the rock medium due to different breakage mechanisms such as crushing, radial cracking and reflection breakage in the presence of a free face. Combined, the crushed zone and radial fracture zone encompass a volume of permanently deformed rock. When the strain wave intensity diminishes to the level where no permanent deformation occurs in the rock mass (i.e. beyond the fragmentation zone), strain waves propagate through the medium in the form of elastic waves, oscillating the particles through which they travel. These waves in the elastic zone are known as ground vibrations, which closely conform to viscoelastic behaviour (Ricker, 1977). The explosion generated waves can be categorized into body waves and surface waves. Body waves travel through the interior of earth and are dilatational or compressive (P-wave) and distortional or shear (S-wave). Surface waves are produced when the spherically radiating body waves impinge on a stress free surface and travel along a surface. It was reported that the surface waves (Rayleigh waves are best known) carry maximum percentage of the radiated energy and will be of significant nature at greater distances from the blast source. Moreover, the frequency of surface waves is lower than body waves and frequently found to be in the range most favourable for structural response (Holloway et al. 1983). All these waves are characterised by an exponential decrease in particle oscillation amplitude as distance from energy source increases (Taqueddin, 1982).

Parameters and Propagation

Whenever blast vibration occurs, it vibrates the ground/soil particles with certain velocity and imparts to it certain amount of acceleration. Ground vibrations are therefore quantified as displacements that very with time accelerations and/or particle velocities at particular ground locations (Duvall and Fogelson, 1962). The passage of viscoelastic waves forces the ground particles to vibrate in an elliptical manner in three dimensions. Therefore, to define the ground motion, three mutually perpendicular components of vibration parameters in longitudinal, transverse and vertical directions are measured. Resultant of all vibration vector components can be calculated from the readings of three mutually perpendicular transducers. However, for industrial purposes only peak values are required. The total time history records are useful for academic purpose.

The propagation of ground vibrations is a complex phenomenon, since rock mass is nonhomogeneous, inelastic and anisotropic. Many researchers, the world over, have correlated the propagation of ground vibrations with different parameters to predict the vibration level. The most widely accepted measurement of ground vibrations considered to cause potential damage to structures is the peak particle velocity ((PPV), defined as the speed at which a particle of earth vibrates as the waves pass a particular section. United States Bureau of Mines has done extensive work in this field and has proposed the following propagation equation, which is valid for many conditions:

$$V = K\,(D/W^{a})^{b} \qquad \ldots.. \qquad (1)$$

Where,

V = peak particle velocity (mm/s),
K = site constant
D = distance between blast site and instrument station (m),
W = weight of explosive used per delay (Kg),
a = exponent, and
b = slope of best fit straight line of V Vs. (D/W^{a}) plot on a long-log scale.

The parameter (D/W^{a}) in the above equation is known as "scaled distance" (SD). Nicholls et al. (1971) suggested that regardless of the blast site, the ground motion component and shot to gauge distance, the SD using square-root of charge weight (a=0.5) provides the most consistent estimate of PPV.

Factors Influencing Ground Vibrations

Charge weight per delay is a very important parameter which controls the intensity of ground vibrations. The intensity of vibrations increases as the quantity of charge detonated per delay increases(Table 1).

Table 1: Parameters influencing ground vibrations (Wiss and Linehan, 1978).

Parameters	**Influence on PPV**		
	Significant	Moderately significant	Insignificant
Controllable variables			
Charge weight per delay	X		
Length of delay	X		
Burden and spacing		X	
Stemming amount			X
Type of stemming			X
Charge length and diameter			X
Angle of borehole			X
Direction of initiation		X	
Charge weight per blast			X
Charge depth			X
Bare Vs. covered detonating cord			
Uncontrollable variables			
General surface terrain			
Type and depth of overburden (Confinement)		X	
Wind			x

The selection of suitable delay interval is extremely important in multi-row blasts. Proper burden relief should be provided to each row for effective movement of the burden rock. If the delay between rows is not enough, the front row burden cannot move forward to sufficient distance to provide free face to the next subsequent row to move out. This adds to more confinement of charges in subsequent rows leading to increased ground vibrations (Fig. 2). It was found that the ground vibration levels could be reduced effectively by arranging delays between rows in such a manner so as to separate the wave fronts emanating from corresponding charges avoiding the superimposition of waves (Wei Wo, 1984).

Damage Criteria

Ground vibrations having sufficient energy can cause damage to structures. The damage level depends upon many factors such as type, condition and age of structure, foundation ground, frequency of vibration etc. As the severity of blast vibrations increases the damage done to structures also increases. Following is the list of types of damage caused to buildings in the order of increasing severity:

1. Dust falling from old plaster cracks
2. Extension of old plaster cracks
3. New plaster crack formation
4. Plaster flaking
5. Plaster dropping large areas
6. Formation of cracks in walls
7. Further severe damage to building

As can be observed from the above points the term "damage" therefore, covers a wide range up to collapse of roof from the extension of a hairline crack. Based on the level of disturbance in the building, various researchers have put forward different damage classifications (Table 2).

Table 2 : Comparison of damage classifications in probabilistic study (Siskind et al. 1980).

Description	Study	Classification
Loosening of paint	Threshold	Threshold
Small plaster cracks at joints between construction elements	Dvorak (1962) Edwards and Northwood (1960) Northwood et al. (1963)	
	Minor	Minor
Lengthening of old cracks	Theonen and Windes (1942)	
Loosening and falling of partitions	Minor	Minor
Cracks in masonry around opening near partitions	Dvorak (1962) Edwards and Northwood (1960) Northwood et al. (1963)	
Hairline to 3 mm cracks Fall of loose mortar	Jensen and Rietman (1978) Langefors et al. (1958)	
	Major	Major
	Theonen and Windes (1942)	
Cracks of several mm in walls	Major	Major
Rupture of opening vaults	Dvorak (1962)	
Structural weakening	Edwards and et al. (1963)	
Fall of Masonry (e.g. chimneys)	Northwood et al. (1963)	
Load support ability affected	Langefors et al. (1958)	

The probability study made by Dowding (1985) revealed that no cosmetic or threshold cracking takes place below a particle velocity of 12 mm/s (0.5 in/s). The data considered by him for the study included the low frequencies of below 4 Hz collected by Dvorak (1962). Residential structures typically resonate at frequencies in the range of 3 Hz to 8 Hz indicating a problem. However, the above study indicated no danger even upto 12 mm/s PPV with such low frequencies. To substantiate this Siskind et al. (1981) conducted studies and observed no blast induced cracking upto 19 mm/s (0.75 in/s) PPV. The probabilistic data provided above may not be valid in specific cases as total time history of vibration event is more important for response spectrum of the structure. Though there is evidence of fatigue effect on blast induced cracking, the effect in general does not appear large in relation to normal blasting activity (Dowding, 1985). The studies made by Theonen and Windes (1942), Leigh (1974), Keowen (1981) and Stagg et al.(1984) clearly indicated the insignificant effect of repeated blasting on cracking. Architects of small House Service Bureau of USA have listed around 40 factors effecting plaster cracking other than blast induced ground vibrations, such as type of material used for construction, its quality, perfectness of construction, age of structure, differential thermal expansion, structural overloading, chemical changes in mortar, bricks and plaster, differential foundation settlement etc. studies made by Holmberg et al. (1981) and Wiss and Nicholls (1974) indicated that the formation and extension of cracks was a function of time and thermal variations respectively.

Human Response to Ground Vibrations

People tend to complain about ground vibrations even below the accepted damage levels because of many reasons. The threshold of complaint for an individual depends upon his health, fear of damage, relationship with mine management, frequency of blasting, duration of vibration etc. As a matter of feet, the human perception of ground vibrations is as low as 1 mm/s (Fig. 3). It is reported that if the local people's attitude is hostile towards mine operators, the tolerance level could be well below 5 mm/s, much below the accepted level of damage.

Damage Levels

Various researchers the world over tried to correlate the damage levels with different parameters of vibration. However, recent investigations have established the peak particle velocity (PPV) concept as the most suitable one for estimating damage levels. Nicholls et al.(1971) established that a PPV below 50 mm/s has roughly 5% probability of causing structural damage. The Indian Standards Institute recommended the following damage criteria (Table 3) without any reference to the condition and state of repair of the structure.

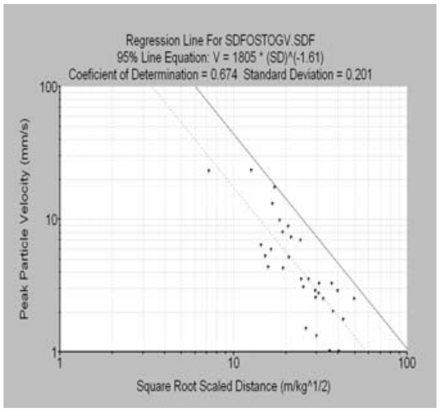
Table 3: Damage criteria according to the ISI (Anon, 1973)

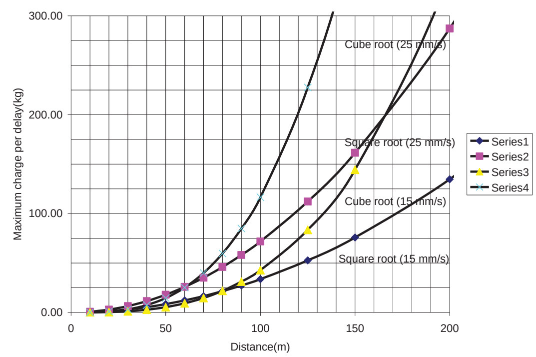
Ground	**Peak particle velocity (mm/s)**	
	From empirical equation	Monitored by instrument
Soil, weathered or soft rock	50	70
Hard rock	70	100

Siskind et al. (1980) recommended that the blast vibration level must be kept below 50 mm/s where frequency is above 40 Hz to minimize the damage. When frequency is below 40 Hz, they recommended that the vibration levels should be kept below 18 mm/s for homes of modern construction and 12.5 mm/s for older homes.

Instrumentation

Blastmate, Minimate are the most commonly used equipment for ground vibration measurement. Use of accelerometers and high frequency geophones is recommended when the expected range of ground vibrations are beyond 254 mm/s which is the maximum range of conventional seismographs. Some instruments used for monitoring ground vibrations are shown below:





Ground vibration measurement Suggested maximum charge per delay

Summary of a typical ground vibration analysis to protect a structure

<table>
<tr><th>Parameter</th><th colspan="2">Range</th><th colspan="2">Class Interval</th><th>No. of events</th><th>Remarks</th></tr>
<tr><td rowspan="3">W
Max.charge per delay (kg)</td><td colspan="2" rowspan="3">15-58</td><td colspan="2">0-20</td><td>18</td><td rowspan="3">Majority of the events (52) utilized less than 40 kg of explosive per delay</td></tr>
<tr><td colspan="2">20-40</td><td>28</td></tr>
<tr><td colspan="2">40-60</td><td>6</td></tr>
<tr><td rowspan="5">PPV,(peak vector sum)
Peak Particle Velocity (mm/s)</td><td colspan="2" rowspan="5">1.11-43.5</td><td colspan="2">5 or below</td><td>36</td><td rowspan="5">36 events have shown a PPV less than 5 mm/s</td></tr>
<tr><td colspan="2">5-10</td><td>8</td></tr>
<tr><td colspan="2">10-20</td><td>4</td></tr>
<tr><td colspan="2">20-30</td><td>3</td></tr>
<tr><td colspan="2">Above 30</td><td>1</td></tr>
<tr><td rowspan="6">Time of occurrence of Peak vector sum(PVS) (ms)</td><td colspan="2" rowspan="6">1 to 288 ms</td><td colspan="2">< 50</td><td>3</td><td rowspan="6">Majority of events recorded pvs within a duration of 200-300 ms.</td></tr>
<tr><td colspan="2">50-100</td><td>11</td></tr>
<tr><td colspan="2">100-150</td><td>9</td></tr>
<tr><td colspan="2">150-200</td><td>5</td></tr>
<tr><td colspan="2">200-300</td><td>13</td></tr>
<tr><td colspan="2">Above 300</td><td>11</td></tr>
<tr><td>Distance (m)</td><td colspan="2">41-274</td><td colspan="3">Blast site to monitoring station</td><td></td></tr>
<tr><td rowspan="3">Frequency (Hz)</td><td colspan="5">Transverse, Vertical, and Longitudinal</td><td rowspan="3">More than 90 % events recorded less than 10 Hz Z_c (Zero crossing) frequency. This has a bearing on fixing allowable PPV.</td></tr>
<tr><td>< 10</td><td>80</td><td>20-30</td><td>31</td><td>40-50</td><td>4</td></tr>
<tr><td>10-20</td><td>22</td><td>30-40</td><td>16</td><td>Above 50</td><td>3</td></tr>
<tr><td>AOP (Air over pressure) , dBL</td><td colspan="5"></td><td>All the events are below the safe limits of 140 dBL</td></tr>
</table>

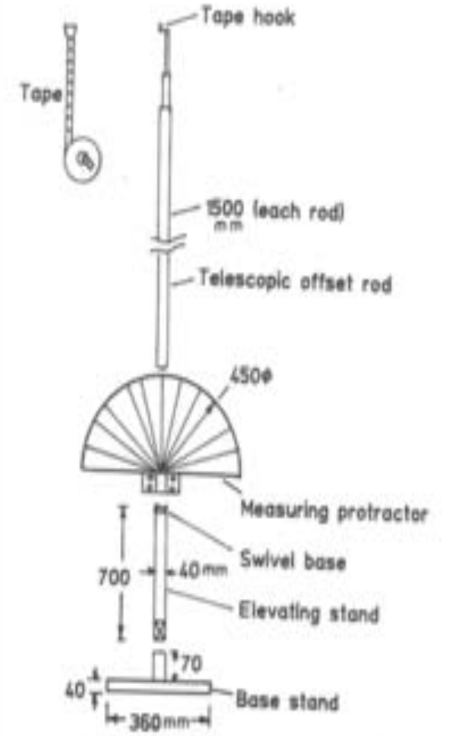


Figure 1- Telescopic overbreak measuring rod

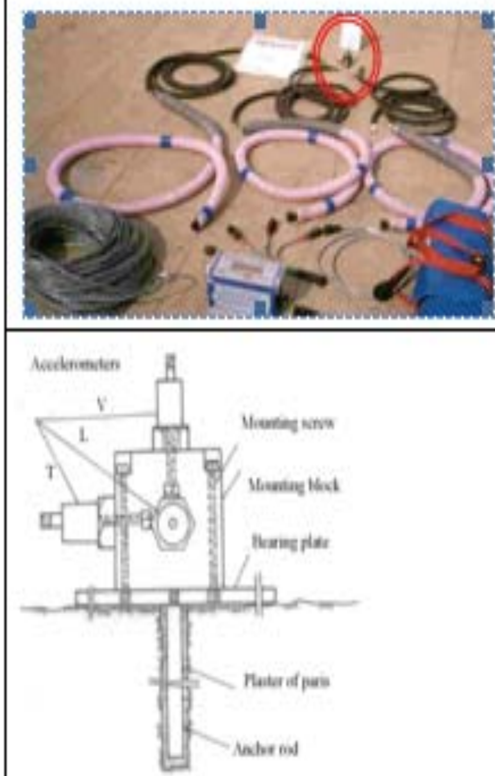
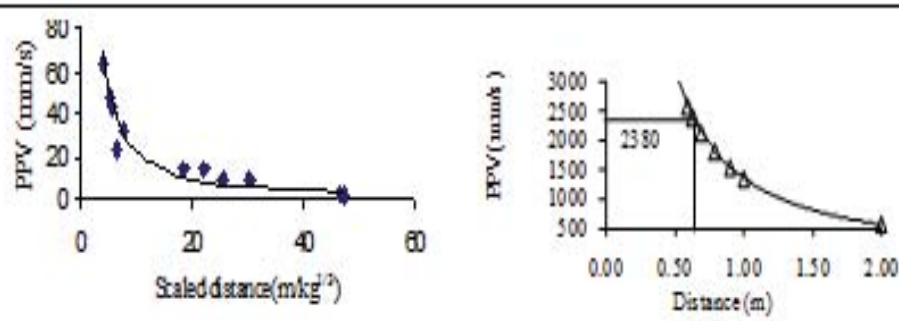


Figure 2 – Fixing of sensors of the accelerometer



(A)

Figure 3(i) PPV predictor for Site-1 and extrapolation upto overbreak distance

Underground blast vibration monitoring-for overbreak control

TYPICAL WAVE FORMS, FREQUENCY,OSMRE AND FFT ANALYSIS

Instantel

Event Report

Date/Time Vert at 12:56:13 February 1, 2007
Trigger Source Geo: 3.00 mm/s, Mic: 238 pa.(L)
Range Geo : 127 mm/s
Record Time 1.0 sec at 1024 sps
Notes
Location: VPT, 0008
Client: , Visakhapatnam
User Name: VMSR Murthy
Converted: February 8, 2007 11:24:58 (V7.01)

Serial Number 4375 V 2.61 MiniMate
Battery Level 6.5 Volts
Unit Calibration August 29, 2006 by CMRI, Dhanbad, Jhar
File Name F375BIZV.XP0

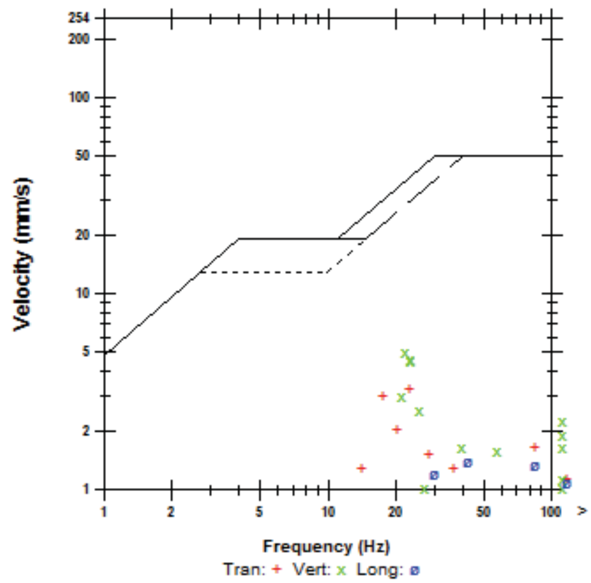
Extended Notes
Ground Vibration monitoring due to underwater blasting at Visakhapatnam Port.

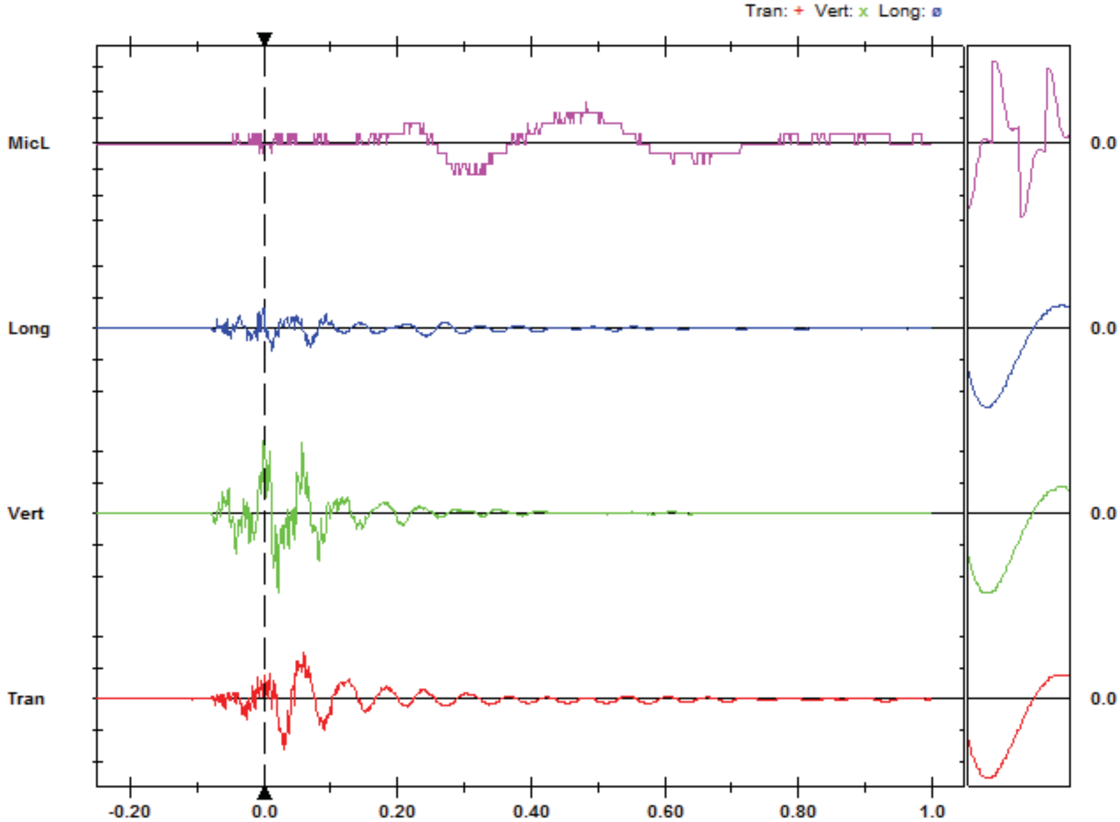
Microphone Linear Weighting
PSPL 8.00 pa.(L) at 0.483 sec
ZC Freq 3.0 Hz
Channel Test Passed (Freq = 20.0 Hz Amp = 451 mv)

	Tran	Vert	Long	
PPV	3.24	5.08	1.40	mm/s
ZC Freq	23	22	47	Hz
Time (Rel. to Trig)	0.032	0.022	0.014	sec
Peak Acceleration	0.159	0.239	0.106	g
Peak Displacement	0.0245	0.0270	0.00701	mm
Sensor Check	Passed	Passed	Passed	
Frequency	7.8	8.0	7.8	Hz
Overswing Ratio	3.7	3.4	3.7	

Peak Vector Sum 5.25 mm/s at 0.022 sec

USBM RI8507 And OSMRE





Time Scale: 0.10 sec/div **Amplitude Scale:** Geo: 2.00 mm/s/div Mic: 5.00 pa.(L)/div Sensor Check

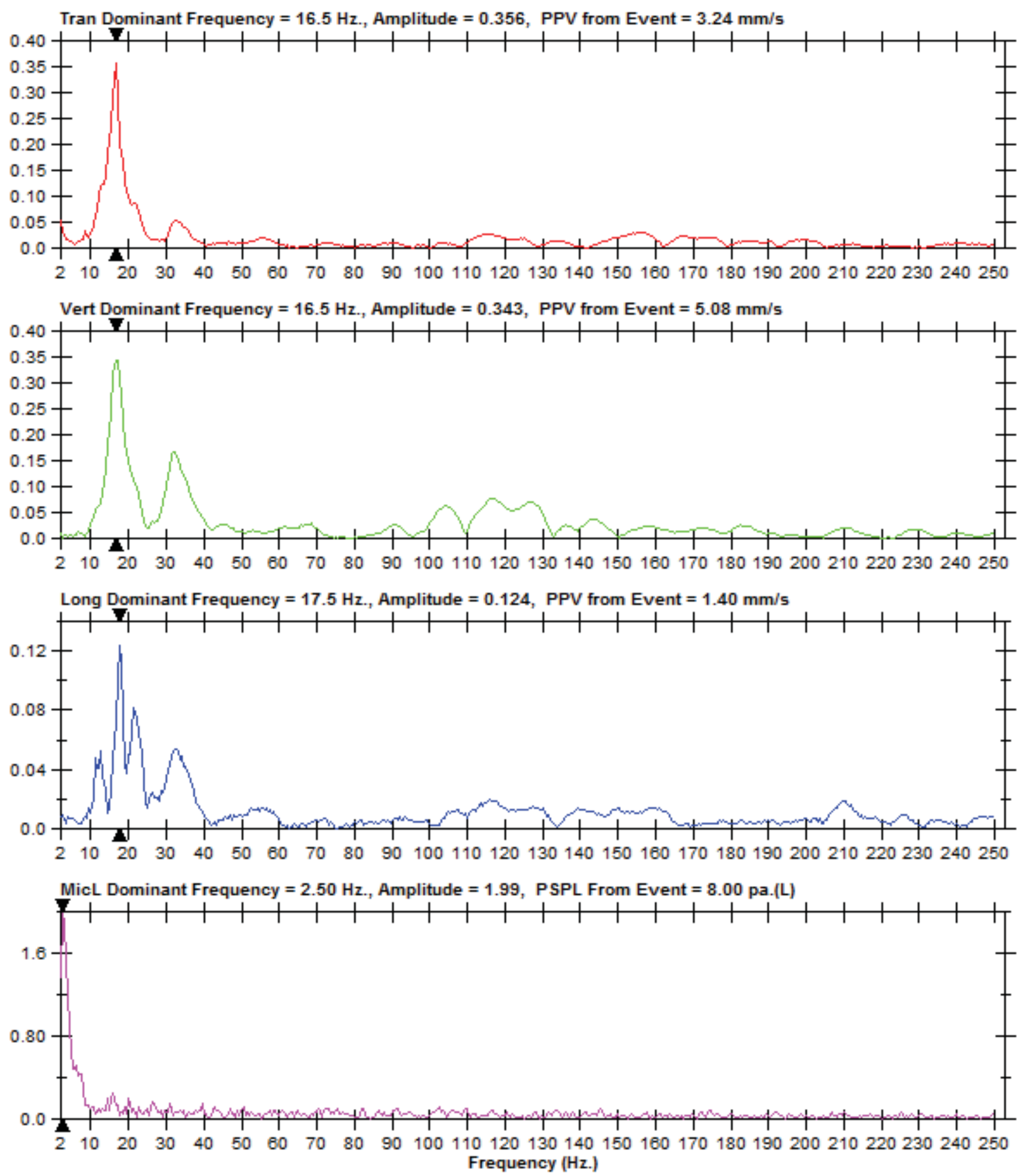
Instantel

FFT Report

Date/Time	Vert at 12:56:13 February 1, 2007	**Serial Number**	4375 V 2.61 MiniMate
Trigger Source	Geo: 3.00 mm/s, Mic: 238 pa.(L)	**Battery Level**	6.5 Volts
Range	Geo : 127 mm/s	**Unit Calibration**	August 29, 2006 by CMRI, Dhanbad, Jhar
Record Time	1.0 sec at 1024 sps	**File Name**	F375BIZV.XP0

Notes
Location: JOB # 0008
Client: VPT, L, Visakhapatnam
User Name: R Murthy
Converted: February 8, 2007 11:24:58 (V7.01)

Extended Notes
Ground Vibration monitoring due to underwater blasting at Visakhapatnam Port.



P4: MEASUREMENT OF VOD OF EXPLOSIVE AND ANALYSIS

Aim: Measurement of velocity of detonation of explosives and its analysis for its performance assessment

Introduction

Real performance of an explosive can be determined by conducting continuous velocity of detonation (VOD) measurements in full scale blast environments. This paper presents and discusses the results of continuous VOD measurements of five different blasting agents carried out at three mines in Turkey. The results show that VOD of a blasting agent varies depending on mainly the rock properties (confinement), presence or absence of sensitizer (chemical formulation), blasthole diameter, and type of primer. The method of continuous VOD measurement is an indispensable and invaluable tool in explosive and primer selection.

VOD measuring techniques

Detonation velocity is an important property to consider when rating an explosive. It has been observed that most of the manufacturers and the utilities rely on calculations based the chemical composition of the bulk explosive to arrive at the VOD value. Needless to mention such interpretation can be fairly inaccurate due to variations in raw material quality, manufacturing process followed and more importantly it is still a theoretical value only. Deriving the correct value of VOD with the proper measurement techniques will result in reduction in the consumption of explosives with optimized results. Presently the explosives used can be in two forms Cartridge & Bulk. Cartridge is a packed form in tubular shape while bulk is free flowing slurry which is put in the hole from a pump truck.

Detonation velocity may be expressed as a confined or unconfined value and is normally given in meter per second (m/s). The confined detonation velocity measures the speed at which the detonation wave travels through a column of explosive within a borehole or other confined space. The unconfined velocity indicates this rate when the explosive is detonated in the open. Explosives generally are used under some degree of confinement. There are number of VOD measuring techniques are available worldwide, each possessing it's advantages and disadvantages. In this paper various techniques are discussed, comparative study is given and attempt has been made to find a cost effective, easy to operate, reliable and accurate method of measuring VOD of bulk explosive in a hole. Based on the method adapted an instrument would be designed and tested. Existing practices for measurement of VOD adapted worldwide can be broadly classified into methods listed as follows:

a) Dautriche Method - In this method two detonation waves propagating from both the ends of explosive column via a detonating fuse bound on an aluminium plate collide. The distance of collision mark from mid-point is measured. This distance is directly proportional to Velocity of Detonation. This method of VOD measurement is suitable for unconfined space where the explosives are used in cartridge form.

b) Photographic Method - Another method in the category is photographic method, where detonation wave is monitored continuously using Streak and framing (high speed) camera. Explosion is an auto luminous process. The light emitted is captured continuously in real time. The VOD can be calculated from the motion video.

c) Discrete points (Point to point) Electric Method – Point to point VOD systems are basically supported by electronics start and stop timer. The one end of sensor cables are inserted into the explosive column at varying distance and other end to the VOD recorder where the start and stop signals are recorded. When detonation reaches the initial sensor timing clock is started and the following sensor cable send the stop signals when detonation reaches to it. The distance between sensor cables are known and thus VOD can be calculated. This method is limited in providing information for critical experiments because of discontinuity of the sensor cable in the explosive column.

d)Resistance Wire Continuous VOD method – This method was developed in early 1960s by the United States Bureau of Mines (USBM) This method basically follows the Ohm's Law(E=RI) where E=Voltage, R=Resistance and I=Current. In this method ionization caused by explosion provides electric short continuously causing the voltage drop monitored by the instrument which is equivalent to the change in resistance value and the constant current. Thus, a voltage drop can be measured instantaneously at any point in time. In this method single wire or two twisted wires of known resistance can be acted as single or double sensors respectively at the same time. It is observed in the experimentation that if in this method sensors are not ruggedized or distance with the return path is not maintained properly for shorting during detonation then the results are usually undeciphered or no readings are obtained.

e) SLIFER continuous VOD system – The SLIFER (Shorted Location Indication by Frequency of Electrical Resonance) system was originally developed by Sandia National laboratories to measure the propagation of shock waves from nuclear explosion. It consists of a shorted length of coaxial cable as a sensor in the explosive column which is connected to the oscillator circuit. This small device shows the frequency which is controlled by length of sensor in the explosive column. As the wire length decreases frequency of oscillation increases. By monitoring this frequency as a function of time, the rate of cable length change can be determined, leading directly to the measurement of VOD. This system has been limited to laboratory work. The restriction with the SLIFER system is that recording cable length with the oscillator is 66m per channel, moreover each sensor must have oscillator connected in a line which should be placed close to the hole or shot area.

f) TDR continuous VOD system – The TDR system originally developed by the Los Alamos National Laboratory to test and verify nuclear reaction yields and stress velocities into the surrounding medium. This system can also be called as CORRTEX system later changed to the VODR-I when the system was declassified for commercialization. In this system narrow electric pulse is sent through the cable sensor and return or reflected path is detected which is from the 262 other end of sensor. This method does not require the sensing cable to be shorted in order to acquire date. This is one of the safest to use with any commercial or military explosive.

g) Method based on fiber-optic – In this method optical fiber is used which is capable of detecting and transmitting a light signal accompanying the detonation wave front. This method is point to point to type wherein the first cable signals the start whereas the second cable placed at a known fixed distance stops the timing clock. The fixed distance between probes divided by the timed clock directly gives the VOD value.

Testing of VOD of explosives by Hand trap

Tape the shorted end of the 30 m (100 ft.) long spool of VOD PROBECABLE-HT to a rock and lower it to the bottom of the hole. Connect the other end of VOD PROBECABLE-HT to coaxial cable by twisting the bare wires and insulating with electrical tape. Load the blasthole with explosives as per normal procedure. Run coaxial cable up to 101m (333 ft.) and plug into the BNC connector on the HandiTrap II™ VOD Recorder. Turn ON the HandiTrap II™ VOD Recorder, press the TEST button to test the VOD circuit, and press the START button. Retreat from the blasting area and fire the blast at any time. The HandiTrap II™ VOD Recorder will record the explosives performance in the blasthole automatically without operator assistance. After the blast, download the VOD data from the HandiTrap II™ VOD Recorder to a PC and view the graph. Point and click to zoom in on the graph and point and click on the graph to calculate and display the VODs anywhere along the explosive's column. Contact MREL to request a link to download a presentation showing a variety of typical VOD results from augured, pumped, cartridge, and decked explosives in dry and wet blastholes.

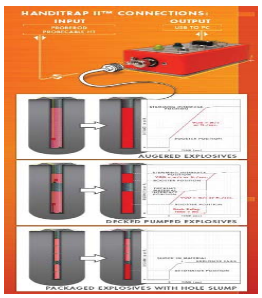


Fig.1 – VOD using Handitrap II (typical)

Fig.2 – VOD using VOD Mate, Instantel

Measurement of Velocity of detonation in field

The continuous VOD measured in one of the holes in the last row of a blast has shown a velocity of 3578m/s which is reasonably good for generating blasthole pressure. This needs to be done on a regular basis for having proper quality checks as well as bring needful changes if any, considering the rock compactness. The weighted P-wave velocity in the bench was found to be 2000m/s. The average density of the rock is 1.934 gm/cc. The effective density of the explosive used was 1.10gm/cc. Impedance matching is done to ensure adequate energy transfer to rock for effective fragmentation. Impedance is computed using the following equation:

$$\rho_e \times VOD \geq \rho_r \times \qquad (1)$$

$$3578 \times 1.1 \geq 2000 \times 1.934$$

$$3936 \geq 3868$$

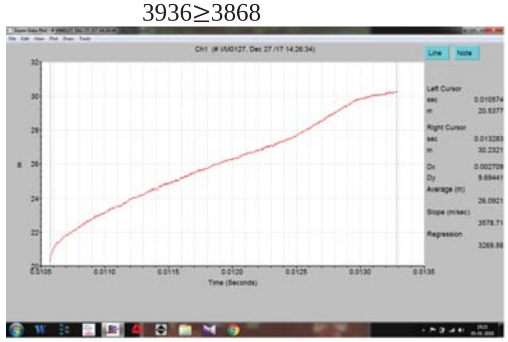


Fig.3- Measured velocity of detonation of an explosive

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P5: FRAGMENTATION AND CHIP SIZE ANALYSIS

INTRODUCTION

Quick and accurate measurements of size distribution are essential for managing fragmented rock and other materials. Various fragmentation measurement techniques are available and used by industry/researchers but most of the methods are time consuming and not precise. WipFrag/Fragalyst is an automated image based granulometry system that uses digital image analysis of rock photographs and video tape images to determine grain size distributions. WipFrag/Fragalyst images can be digitized from fixed video cameras in the field, or using roving camcorders. Photographic images can be digitized from slides, prints or negatives, using a desktop copy stand. Digital images in a variety of formats, delivered on disk or over electronic networks, can be used.

Fragmentation measurement techniques

Blast optimization requires a degree of compromise between the competing objectives of maximum fragmentation, minimum dilution and minimum costs for drilling and explosives. Also, mining companies and quarry operations have to examine and reduce production costs to remain competitive. But no single factor, such as cost of explosives, can be properly evaluated without measurements of fragmentation and rock quality. Hence the need to manage production costs necessitates the need to measure the post-blast fragmentation. Quantification of fragmentation on a larger scale is an extremely complicated task. Because it needs a substantial amount of time to find out manually the grain size distribution in a muck pile. Research has been carried out worldwide with different methods and tools for measurement of fragmentation. These methods are listed below.

- Sieving or Screening.
- Oversize boulder count method.
- Explosive consumption in secondary blasting method.
- Shovel loading rate method.
- Bridging delays at the crusher method.
- Visual analysis method.
- Photographic or manual analysis method.
- Conventional and high-speed photogrammetric method.
- High speed photography or image analysis method.

WipFrag image analysis software

The WipFrag image analysis software is recently developed granulometry software (Maerz et al., 1996). It utilizes the technique of measurement of fragmentation with the help of digital images of blasted muck pile. Usually camcorder images of the muck pile are captured in the field. A Cannon or Sony Hi-8 video camcorder is offered as a standard for this purpose. These images are then transferred to the WipFrag system. The WipFrag system consist a software pack installed in a computer. A hard lock key is provided with the WipFrag software pack as security equipment which has to be inserted in the CPU to run the software. The images of the muck pile transferred to the WipFrag system are analysed by delineating the fragment edges with automatic netting followed by manual editing. These Network areas are converted into volumes and weights and the resulting data is displayed as a graph. The detailed methodology of fragmentation analysis with a WipFrag system is discussed in further sections.

Rock pile sampling and Photography (Franklin et al., 1996)

Rock pile sampling refers to the process of collection of the blasted muck pile samples in the form of digital images by capturing photographs. The basic underlying rule in sampling is that the larger the number of images processed, the more reliable the results. This is due to the non-homogeneous nature of muck pile and the greater statistical sampling base. Rock sampling is a complex process involving following steps –

- Selection of suitable viewpoints from where the most representative rock pile samples can be collected. Since we can only measure what we can see, so to obtain reliable results of the fragmentation analysis some precautions have to be taken during selecting viewpoints.
- At least one scaling object of known length has to be positioned near the edge of the muck pile so as not to obscure the rock we are trying to measure.
- A suitable camcorder is then set in front of the muck pile and several shots are taken. Preferably at least five shots are taken at random locations for a large rock pile. For improved estimates of oversize the number of full-scale shots should be increased to at least ten.

Precautions

Following precautions have to be taken during photography –

- Avoid wide-angle close-up photography and oblique shots that distort the scale. If the rock pile surface is oblique to the camera, place identical scaling object at the nearest and the furthest points that can be averaged or used in auto-tilt correction.
- Since the WipFrag detects the rock edges depending on the intensity of light, provide uniform indirect or diffusing light without excessive sharp or one-sided shadows and “hot spots” for photography. WipFrag works best when each fragment is equally bright and surrounded by a thin, uniform shadow.
- Choose dull days in preference to bright sunlight for photography.
- Beware of rock pile segregation. Large blocks tend to roll to the outer edges and fines may cover the surface or become hidden as a result of gravity or rainfall. The effects can be minimized by increasing the number of images per sample but only with careful selection of image locations.
- Maintain the camera in good working condition and protect it from dust and mechanical damage.

METHODOLOGY

Transferring the digital images to the WipFrag system

The images of the muck pile captured during rock pile sampling are uploaded into the WipFrag system through a suitable data cable. This data cable connects the camcorder directly to the CPU. The images are saved in the computer hard disk drive as bitmap (.bmp) or jpeg (.jpg) files.

Opening an image in the WipFrag window for analysis

For opening an image of the blasted muckpile in the WipFrag window, the File menu is opened and "OPEN" option is clicked on. This command opens a dialogue box from where a photograph may be selected and opened.

Set tilt option

After opening a muck pile photograph on the WipFrag window, the tilt correction is done. For obtaining precise results, it is necessary during photography that the horizontal axis of the camera should be normal to the surface of the muck pile. But this is practically impossible to keep the horizontal axis normal to the inclined face of the muck pile. Hence the images are captured from the viewpoint exactly in front of the muck pile with the horizontal axis of the camera at some angle to the face of the muck pile. For capturing images of the muck pile in such a manner, the use of a scaling object becomes necessary. Two white scale bars or staffs of known length are suitable for this purpose. One of the two bars is placed at the top edge of the muck pile while the other is placed at the bottom edge. The image captured with the help of these bars will be adequate for tilt correction during analysis. Hence before we start the analysis of the rock pile sample, we have to set the tilt of the muckpile profile. This can be done by clicking on the "Set tilt" option in the "Fragmentation" menu.

Edge detection settings

After the tilt scaling the most important operation to be performed is the Edge detection settings. The edge detection settings include the settings of a set of certain parameters called the edge detection parameters. Edge detection parameters are the numerical values used by the WipFrag during the various stages of fragment edge detection. To perform the edge detection settings the option "Edge detection parameters" in the Options menu is clicked on.

Generating net

Net generation refers to the process of generating the overlay network of block outlines. These outlines delineate the edges of the fragments. The option for generating the net appears as we click on the "Fragmentation" menu on pressing the "Generate net" option the automatic edge detection automatically runs through a series of edge detection operation to obtain a net of lines corresponding closely to fragment boundaries. This process typically takes a few seconds and for this tenure the WipFrag displays a status window showing the progress status of auto netting.

WipFrag Output

The output of the granulometry analysis with the WipFrag software essentially consists of a cumulative size table either in ISO or US standard sieve sizes whichever has been selected during sieve default settings. Image analysis with Wipfrag/Fragalyst software are shown in Figure 1.

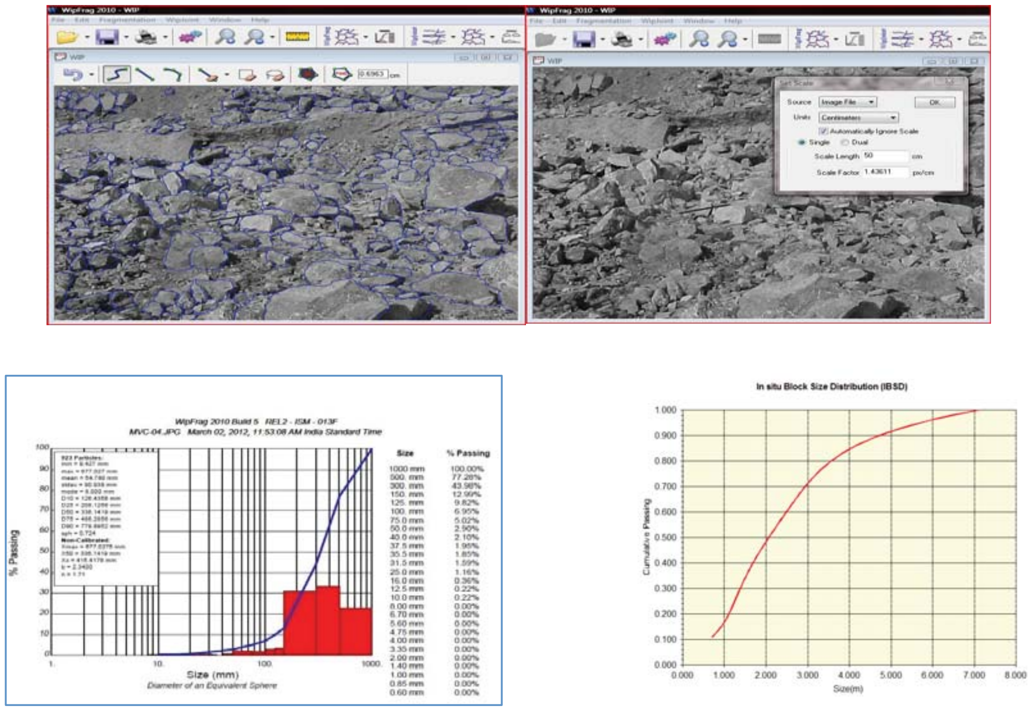

Fig.1: Image analysis process in WipFrag/Fragalyst Software

CONCLUSIONS

- The WipFrag is efficient fragmentation analysis software and it is a direct method of fragmentation assessment as compared to the other methods such as the shovel loading rate method and the explosive consumption in secondary blasting method.
- WipFrag is a quick and time saving assessment technique.
- It possesses a high degree of accuracy and precision
- Multiple image analysis technique is better for optimum rock size distribution than single image analysis WipFrag software.

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P6- BURDEN VELOCITY MEASUREMENT AND ANALYSIS USING HIGH SPEED VIDEO CAMERA

Aim: To determine the burden velocity of blast face for evaluating the blast performance

Introduction

Blasting being a pretty short lived (less than 3 seconds) activity requires high speed video monitoring for optimising blast performance and is a standard practise in many large production mines. The recent development of low-cost, easy-to-use high-speed digital cameras and user-friendly software can help mine management to monitor blasts effectively and reduce costs. Digital high-speed video and image analysis software can accurately quantify each blast. Analysis of the images can facilitate the identification of causes of variance from design, geology changes, errors in blast practice and areas of poor performance. The use of high-speed video greatly enhances the audit and review processes so that future blast design can be quickly and appropriately modified to deliver improved blasting performance and reduced costs. Capturing images at 1000 frames per second (fps) allows analysis on a millisecond timescale, where detonator timings, fragmentation, venting, fume generation and fly rock can be accurately identified. Today, blast analysis is often conducted based upon measurement of the design and the end resulting muck pile. The missing components can be the analysis of the rock mass being blasted and the dynamic process happening in real time within the blast itself. This is best achieved by playing detailed visual information. This method provides scope for understanding the short falls in the system and also provides opportunity to learn

Applications

Seeing is believing. Provides scope for analysing the and interpreting according to particular requirements. Rapid improvement in conventional video cameras, digital high-speed video cameras and user-friendly image analysis software has changed all of that. Today, any company in the extractive industry – small, medium or large – can afford to operate quite advanced high-speed digital imaging systems and image analysis software as an integrated part of its operations. The cost benefits of such procedures are so well established that to not use such systems can place an operation at a distinct disadvantage to its competitors (Giltner and Worsey, 1986). In addition, the increased accountability that extractive industries are facing demands irrefutable proof that good management and environmental procedures are being followed.

The application of modern digital imagery and software can:

- monitor and analyse blasting performance
- help assess crusher performance
- aid dragline, shovel and/or loader performance monitoring
- assess conveyor belt damage analysis while the belt is in use
- record accident impact monitoring on driverless mine vehicles
- record track and wheel condition on moving ore trains
- monitor the cowcatcher impact of driverless ore trains.

Digital Systems in Place

Domestic digital video

Advantage of this system was the immediate replay facility allowing the visual analysis of data whilst the event was still clear in the minds of the operators, thereby maximising the evaluative capabilities of experienced staff.

Domestic digital slow-motion cameras

The emergence of domestic digital slow-motion cameras that can record in HD to about 150 fps has revolutionised the analysis capabilities available because of the higher frame rates these cameras can achieve. They have one major drawback: image compression. All domestic digital cameras compress the image file to allow the fast transfer of images to the recording medium.

Low-end digital high-speed video

This is the area of rapid expansion in both the type and quality of the equipment available. High-speed digital video equipment allows the capture of the dynamics of a fast event where the details of the event – such as blast detonation timing, fragmentation, misfiring, geology changes and blast breakthrough – are sampled and recorded. The event can be viewed or replayed at a speed whereby the human eye can detect all or most of the dynamic action captured. This provides the information necessary for an experienced engineer or technician to evaluate what happened. This procedure is extremely valuable when you are trying to analyse blasts, conveyor belt systems, crushing processes, drilling or any mechanical or dynamic processes that happen faster than the human eye can follow. Once captured, the images can then be downloaded to a computer for replay, analysis and archival purposes.

High-end digital high-speed video

High-speed camera images have always been considered the 'gold standard' when conducting image analysis of blasting and other high-speed processes. The quality of the images is high, and the ability to analyse single images and then compare sequential images has made this an invaluable tool for those who could afford the high costs involved. The high quality of digital high-speed imaging systems were usually too expensive and require trained technicians to operate them.Typical systems reported are given in Table 1.

Table 1. Equipment comparison table.

Product	Resolution and frame rate	Compression	Analysis capable	Approximate cost ($)
Domestic digital video recorder	1920 × 1080 25/50 fps	60-100/1	Partly, restricted by compression and frame rate	1100-3500
Domestic digital slow-motion camera	1920 × 1080 90-150 fps	100/1 or higher	Better analysis but no time code	900-2500
Low-end digital high-speed cameras	1920 × 1080 to 700 fps 1280 × 720 at 1000 fps	Zero compression	Excellent, accurate time code and pixel ID	25 000-40 000
High-end digital high-speed cameras	2560 × 1920 at 2000 fps 1920 × 1080 at 5000 fps 1280 × 720 at 10 000 fps	Zero compression	Excellent, accurate time code and pixel ID	65 000-125 000

(Reference 1)

Purpose :

- Stemming retention studies--effects of changing stemming type and height and the performance of stemming plugs
- Top movement studies including cratering and doming, cutoffs, and fly-rock
- Face movement studies including front row burden, blowouts, hard toe problems, material trajectory, velocity, and casting range
- Timing studies, including detonator delay time quality, actual blast hole
- sequence, and blast delay times achieved, effects of timing changes, cutoffs, and misfires
- Environmental studies including fly-rock, air-blast noise from blowouts, vibration from choked blast, back-break, and noxious fumes

Key Features:

- Handheld and autonomous systems available
- Setup and control via multiple interfaces
- Exceptional resolution and frame rate options
- Advanced synchronization and control of multi-camera systems
- Unique imaging capabilities for greater system magnification & depth of field
- Long record times - onboard mass storage up to 2TB
- Flexible workflows for unique applications

Equipment:

Motion analysis of face and stemming ejection form two crucial parts of high-speed imaging. A typical view of the same is provided in Fig.1.

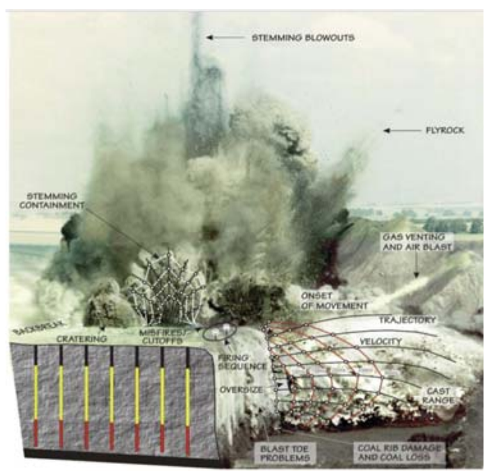

Fig.1 – Blasting impacts and high-speed video application

Typical High-speed video camera developed by MREL being used for blast analysis is presented in Fig.2.The detonation timings being captured is shown in Fig.3. ProAnalyst is the software used for analyzing the blast motions and determining the burden velocities. Burden velocities in different face conditions are critical to evaluate and these range from 20 to 40 m/s. Analysis of these velocities indicate the burden design and delay allocation with respect to their adequacy for achieving proper fragmentation and also control excess throw/fly rock.

Fig.2 - High speed video camera (Courtesy, MREL web site))

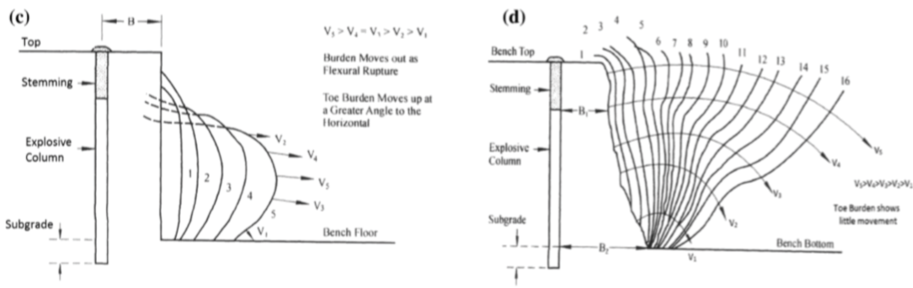

Fig.3 – Burden movement(velocities) in different face conditions (Chiappetta RF, Mammele ME (1987)

Table 2 – Typical delay between rows suggested (Sastry et al., IJGGE,2015)

BH/B Ratio	Avg. Burden Velocity (m/s)	Delay Time Required (ms/m)
2.0	79.8	12.5
2.5	95.6	10.5
3.2	100.7	10.0
3.75	127.9	8.0

Common blasting concerns

Competing factors control blasting practices. Explosives manufacturers would prefer to sell the maximum use of explosives for a blast, mine managers would prefer to buy the least amount of explosives to keep costs as low as possible, while mill operators would like a steady and uniform feed size to maximise throughput. In addition, environmental and community concerns need to be correctly addressed and monitored. Finding the balance between too much and not enough keeps blasting consultants and others in business. Harrington (2005) said: ‘measurement is the first step that leads to control and eventually to improvement. If you can’t measure something, you can’t understand it. If you can’t understand it, you can’t control it. If you can’t control it, you can’t improve it.’

Measurement is usually based around muck pile shape, dig times, overburden removal, mill throughput and the requirement for secondary breakage. Blast designs are altered to improve these parameters. What are not measured in this process are the sources of poor compliance or variance between desired outcome and broken rock on the ground. High-speed video can measure parameters as the fragmentation happens. Using a high frame rate, and provided that the camera is placed strategically so that the critical parts of the blasting process are visible, the video can capture:

- variances in detonator timing
- sources and location of variable fragmentation
- stemming ejection/rifling
- sources of NOX/fume creation
- sources of flyrock
- breakthrough into orebody/coal seam
- face bursts.

Table 2 lists the speeds at which common blasting events happen and suggests some minimum frame rates that could be used to capture detailed information.

Table 2. Blasting event speeds.

Blasting event	Velocity (m/s)	Velocity (m/ms)	Frame rate for 1 m of action/fps
Detonating cord	7000	7	7000
Signal tube	2000	2	2000
ANFO	2500	2.5	2500
Emulsion ANFO blends	3500	3.5	3500

(Reference 1)

There are three interlinked parameters that limit the ability of any digital photographic method. These are shutter speed, aperture and sensitivity of the image chip. The shutter speed can theoretically be as slow as the inverse of the frame rate, eg 1000 fps shutter speed has to be faster or equal to 1/1000 sec. Faster shutter speeds means that the chip receives light in whatever fraction of a second the shutter is open, and either the aperture needs to be opened to allow more light in at the cost of depth of field or the chip sensitivity needs to be increased to react to lower light levels. Improvements in chip technology have allowed for more sensitive chips. This means that we can capture events with fast shutter speeds and reasonable apertures to maintain depth of field. Figure 1 is a still from a modern high-resolution camera with a frame rate of 1000 fps. At this rate, signal tube lengths at around 2 m are illuminated. At this frame rate, each signal tube will be captured in a frame and each detonation will also be displayed. This allows auditing of the design for implementation compliance and enables troubleshooting of design flaws as they will be seen in the playback of the video. Visible in Figure 1 are the signal tubes burning, in each case the length of burn is less than the spacing between holes. A considerable amount of energy is directed upward, and the resultant local flyrock is visible. Typical burden velocities monitored are shown in Fig.4.

Limitations

The use of any visible light technology is limited by the available light. Newer sensors have improved sensitivity so that this is becoming less of an issue. The presence of dust between the camera and the subject of interest can obscure vision; however, appropriate camera placement and mine site dust management procedures will mitigate this. Most of the data collection that may need to be analysed, eg detonator timings, will be captured before generated dust obscures the field of vision. Also, the use of multiple cameras when necessary can alleviate the loss of vision in this event.

As previously discussed, standard video is limited by frame rate; however, ground vibration has a more significant effect on the quality of images collected. By contrast, high-speed cameras have two features that effectively negate ground vibration interference with image quality. These are the frame rate and the high shutter speeds that they offer. The combination of these two parameters means that the camera movement is so slight when the shutter is opened and closed that it is insignificant in its effect on image quality. The arrival of the ground vibration waves will be captured by the camera movement as the centre of the image will change with the movement of the camera and tripod. This can be seen on playback and may be of assistance in ground vibration analysis. Lack of training of camera operators can reduce this technology to a dust-collecting shelf item.

A rudimentary knowledge of photographic techniques needs to be developed to ensure that the camera is set up in the most appropriate available location. In conjunction with this, a working knowledge of the blasting process is required to ensure that the operator can capture the correct subjects of interest. To obtain maximum benefit from the captured images, it is desirable that the initial analysis and communication with the blast crew happens before the crew signs off and leaves the site by reviewing the footage with the drill and blast crew immediately after the event. Important information related to blast performance can be documented whilst it is still fresh.

Summary

By utilising this in blasting, refining blast designs, auditing designs and evaluating good footage, a mine may reduce the explosives consumed and achieve a better result simply by being able to measure and thus control more aspects of the blast process. If an environmental problem exists, visually recording current practices and then visually recording the improvements provides powerful evidence that the improvements made demonstrate due diligence. This helps to remind legislators and courts of the success of the program that the mine implements. It is essential that there is an impartial witness, that is, unedited footage of all that transpired. Being in control of the visual images used in an environmental argument provides the best way to allay fear within the community.

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P7- FRACTURE TOUGHNESS

Aim: Determination of fracture toughness of rocks (tensile mode- Mode I)

Fracture toughness is a property, which describes the ability of a material containing a crack to resist fracture. It is the fundamental property of the material. Fracture toughness is also called the stress intensity factor. If a material has a large value of fracture toughness it will undergo ductile fracture. Brittle fracture is very characteristic of materials with a low fracture toughness value. It implies that higher the value of fracture toughness for a rock more hard it will be to drill in it and vice versa. For determination of the critical stress intensity factors of the different modes, i.e. fracture toughness's K_{IC}, K_{IIC} (and K_{IIIC}), respectively, laboratory testing methods have been developed. Most matured are the Mode I testing methods, evidently in three ISRM Suggested Methods. Some Mode II methods exist but most are insufficient to provide reliable results. There are very few methods available that provide Mode III loading conditions (e.g. Cox & Scholz, 1988; Yacoub-Tokatly et al., 1989). Only Mode I fracture toughness testing methods are summarized below as this test is commonly considered.

TEST METHOD AND PROCEDURE

The fracture toughness determined in accordance with this test method is for the opening mode of loading. There are two methods, to determine the fracture toughness in the laboratory (Single edge notched round bar and Chevron edge notched round bar). Figure 1 shows the specifications of Sample required for both the tests method.

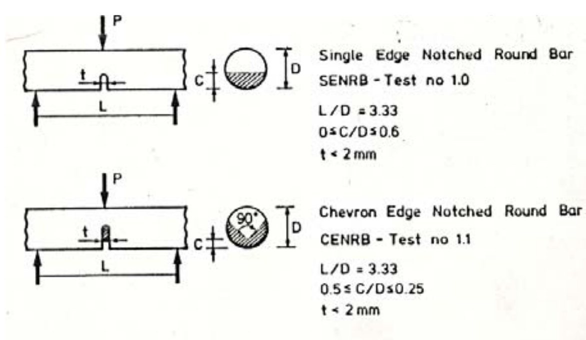

Fig. 1: Specifications of Sample Req. for FT tests

a) The diameter (D) of the core sample is measured. The depth of the notch is marked on the sample such that it lies between 0.25 and 0.5 times the diameter of core. The projection of the notch line must pass through the center of the core and prepared a V-shaped notch perpendicular to the core axis as shown in Figure 2.

Fig. 2: Prepared notch in rock sample for fracture toughness test

b) The sample is placed on the testing apparatus as shown in Figure 3. Care is taken before applying the load that sample must touch the support rollers and placed equidistant. The loading roller must lied vertically above the notch.

Fig.3: Test apparatus for fracture toughness test

c) Load is gradually applied till the sample breaks along the notch as shown in Figure 4. The applied breaking load is measured in MPa. A view of the broken samples along the notch is shown in Figure 5.

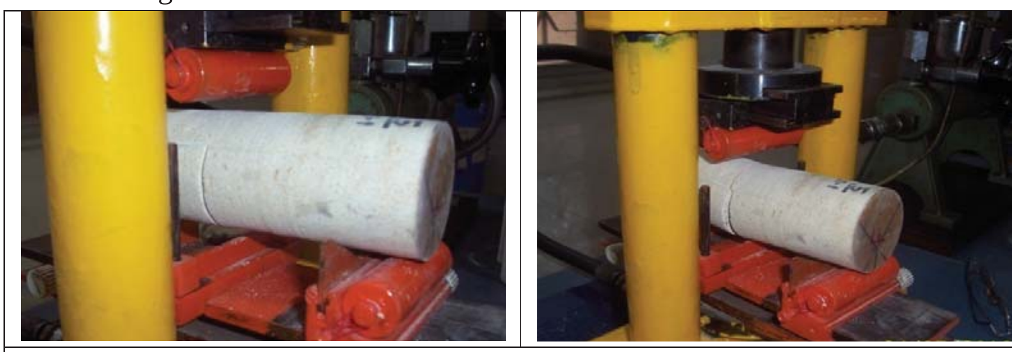

Fig. 4: Status of sample before and after applied load

Fig. 5: View of broken samples along notch

CALCULATIONS

The fracture toughness (K_{IC}) for SENRB-geometry is calculated as:

$$K_{IC} = \frac{10.617\left(\frac{C}{D} + 19.646\left(\frac{C}{D}\right)^{5.5}\right)^{0.5} P.}{\left(1 - \frac{C}{D}\right)^{0.25} D^{0.5}}$$

Where,

C = depth of notch (mm),

D = specimen diameter (mm) and

P = load (MPa).

The formula for CENRB is as follows:

$$\boldsymbol{K_{IC} = 0.8325\left(7.2984 + 540.26\frac{C}{D} - 122.34\left(\frac{C}{D}\right)^{2} + 374.67(\frac{C}{D})^{\wedge}3\right) \quad P/D^{1.5}}$$

Where,

C = depth of notch (mm),

D = specimen diameter (mm) and

P = load (MPa).

Because of the low loads needed in fracture toughness testing the procedure of using the pressure manometer for pressure reading only will give a rough estimate of K_{IC}. The accuracy can be improved by using low range manometer. The load is applied to the specimen until it falls. The maximum load is given by,

P = 800* P (N)

Where,

P is the maximum pressure in MPa.

K_{IC} evaluated according to the formulas give above. As the notch prepared will not have an accurate dimension, some corrections need to be applied to the calculated FT value and Corrected FT was obtained (Table 1).

Table 1: Showing the values of FT, Corrected FT SJ, Brittleness Value and DRI

Diameter	Notch Angle	Depth of Notch	Failure Load	S-J Value	Brittleness Value	DRI	FT	Corrected FT
47.38	1.971	0.98	2.20	69.79	35.2611321	46	0.030	0.026
47.33	2.298	1.70	1.70	119.56	48.0280791	63	0.022	0.016
47.36	1.814	1.00	1.50	117.63	20.5919112	35	0.018	0.015
47.41	1.989	4.7	2.80	127.16	31.2662911	47	0.029	0.020
47.48	1.980	6.11	1.40	129.44	23.7888968	41	0.017	0.013
47.37	1.928	7.67	1.80	94.54	51.3695264	63	0.025	0.022
50.67	2.006	5.19	1.90	83.51	41.5950943	51	0.018	0.013
47.44	1.788	7.33	2.40	79.58	26.3784971	36	0.032	0.029
47.34	1.989	6.44	1.80	92.99	18.5632466	28	0.022	0.018
47.41	1.579	8.04	0.98	109.60	51.8525802	69	0.014	0.014
47.39	1.928	8.07	1.03	134.75	64.7416541	85	0.014	0.013

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P8: CERCHAR HARDNESS AND CERCHAR ABRASIVITY INDICES

Aim: Determination of Cerchar hardness and Cerchar abrasivity of rocks

A) CERCHAR HARDNESS INDEX

Cerchar Hardness Index represents the degree of resistance offered for penetration due to surface hardness of rock specimen. Its use is found mainly in selecting suitable cutting tools, determining drill specifications and selecting excavation machinery etc.

TEST PROCEDURE

In this method a test bit having a dia of 8.5mm is allowed to penetrate under a static load of 70N/200N/400N depending on the strength and penetration of rock. The rpm of the bit is 200. The depth of penetration is recorded as a function of time. A LVDT is placed to measure the penetration of bit with time through a data acquisition system. Subsequently, the penetration history is analyzed for the slope of the curve and then converted to CHI from a calibration graph developed. The test setup is shown in Fig.1. A cuttability classification for Cerchar hardness as proposed by Bilgin, 1992 is given in Table 1. The bit specifications are shown in Fig.2 and the weights used for varying the thrust (200 N and 400 N including the spindle weight) are shown in Fig.3.

Table 1: Cuttability classification based on Cerchar hardness index (Bilgin et al.1992)

CHI	Cuttability
< 15	**Easy**
15-21	Normal
21-54	Difficult
>54	Very Difficult

Fig. 1: Cerchar hardness testing machine

Fig.2: Miniature drill machine) with bit specifications

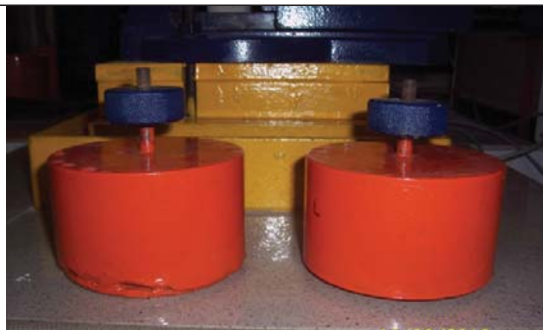	
Applied load (6.5 kg each)	Applied load (16.5 kg each)

Fig. 3: Applied load for miniature drill test

Typical test results of some test samples are provided in Table 2.

Table 2: Cerchar Hardness Index Values

Borehole	Cerchar Hardness Index
BH No.1	35,6, 42.6, 55.9
	24.6, 36.7, 99.8
	14.3, 16.7, 91.8
	77.6, 89.5, 103.8
	22,27,46
BH No.2	60.2, 65.4, 78.7
	44.4, 50.5, 56.4,
	32.2, 44.8, 54.2,
	65.3, 82, 86.7
	30.6, 35.7, 45.3
	32, 48,62
BH No.4	18,28,66
	22.4, 30.4, 35.5
	52.8, 67.3, 72.4

Typical Penetration CHI Signatures of Rocks

Some typical signatures of the above tests are provided in Fig.4 for ready reference. The CHI variation was observed in some of the samples mainly due to the mineralogical variations. The test results varied in some of the rock suits probably for the following reasons:

- Some of the samples are relatively harder due to variation in silica percentage as well as due to the presence of hard minerals thus resulting in higher CHI values.
- From physical examination it is also observed that some of the samples have undergone some changes in their grain texture and structure due to metamorphism of rocks.
- The grain size was coarser in general and finer in some samples. Fine grain matrix coupled with alignment features present indicates the onset of metamorphism. Coarser are relatively softer in general.
- Where penetration is more than 50 Points in the CHI scale the rocks are grouped into hard category and the standard method is to use 400 N load for determining the Cerchar Hardness Index. The CHI values are the average of 3 tests carried out on each rock sample.
- The penetration rate (depth vs time) in case of sample no.31b was very high thus there is a sharp increase in a short time. This is indicative of a soft category rock. It may be noted that the penetration achieved in this case was with a standard load of 200N. This is indicative of a medium hardness category.
- Whereas the penetration rates in case of sample no.s 73b and 16 were with a vertical load of 400 N because of the extremely lower penetration achieved with 200 N standard load. This is indicative of hard rock. The general values are also observed to be high in the range of 27 to 78.7 excepting one value around 22. This could be attributed to localized mineralogical variation.
- The graph in red colour is the actual penetration with time whereas the graph in blue colour is the scaled penetration for computing the Cerchar Hardness directly using a data acquisition system.

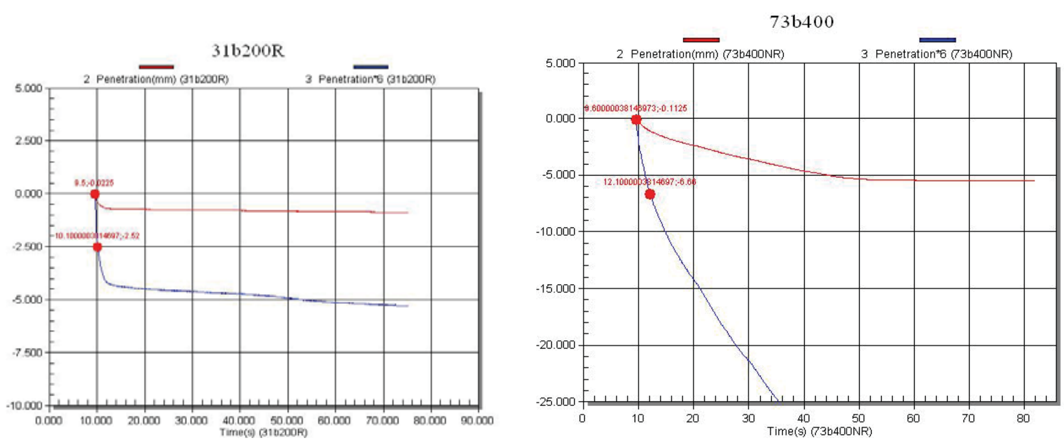

Fig. 4: Test signatures of CHI for some samples

The variations in the results, as observed in some of the samples, are due to the sample location(x,y), depth(z), degree of recrystallisation the sample undergone, grain size, mineralogy(a major factor), cementing material, soft/hard mineral intrusions. It is suggested to consider the average of these three values for CHI though the highest values have to be taken into cognizance for critical system designs.

B) CERCHAR ABRASIVITY INDEX

Scope: The CERCHAR Abrasivity Test is intended as an index test for classifying the abrasivity of a rock material.

Apparatus:

1. Stylus: It is strongly recommended to employ only styli tempered to Rockwell hardness HRC 55 ± 1. The diameter of the stylus should be at least 6 mm. The tip of a stylus shall have a conical angle of 90°. A worn stylus should be re-sharpened and the tip angle checked under a microscope before use in a further test.
2. Force: The static force acting on the stylus should be 70 N.
3. Grinder: Each used stylus should be re-sharpened using a standard abrasive stone wheel.
4. Test Specimen: The rock sample may be either disc-shaped or irregular in shape. The size of the rock surface should be sufficient to permit five test scratches that are at least 5 mm from the edge of the rock surface. Each test should be 5 mm apart.

Test Procedure:

1. The sample should be clamped firmly in the vice while observing the desired scratching direction.
2. The rock surface should be, to the extent possible, horizontal.
3. The stylus should be carefully lowered onto the rock surface to avoid any damage to the tip of the stylus.
4. The stylus should be positioned so it is vertical and perpendicular to the rock surface.
5. The length of a test scratch in the rock sample must be exactly 10.0 mm within 1 s.
6. During the test there should be constant contact between the stylus and the rock surface.
7. After testing, the stylus is carefully lifted from the rock surface and the stylus removed.
8. A minimum of five test replications must be made on the rock surface, each time by a new or re-sharpened stylus.

Stylus Wear Measurement:

1. The length or diameter of the wear flat, d, shall be based on optical and digital methods using a microscope having a minimum magnification of 25X.
2. The measuring resolution should be at least ±-0.005 mm with readings reported to the nearest 0.01 mm.
3. When using the side-view method, it is suggested the stylus should be placed in a V-notch holder or jig and four measurements shall be made each at 90° rotation.
4. The measurements should be taken parallel and perpendicular to the direction of scratching.

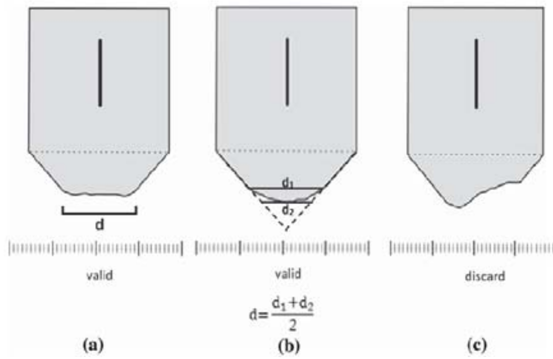

Fig. Standard worn profiles (a, b) and the corresponding length of wear surface, (c) an example of a non-standard profile in which case no measurement should be recorded.

Calculations:

For each measurement of the wear flat, d, the CAI is calculated by the formula given in Eq. (1)

$$\mathbf{CAI = d \times 10} \qquad \mathbf{(1)}$$

where d is the wear tip surface measured to an accuracy of 0.01 mm.

The dimensionless CAI value is reported as the arithmetic mean of five or more test replications.

Classification of CAI:

Mean CAI	Classification
0.1–0.4	Extremely low
0.5–0.9	Very low
1.0–1.9	Low
2.0–2.9	Medium
3.0–3.9	High
4.0–4.9	Very high
≥5	Extremely high

Data Table:

SN.	Rock Sample	d_1	d_2	d_3	d_4	Avg. dia (d)
1						

Results:

Precaution:

1. Height of sample should be enough to keep the loading axis perfectly vertical so that eccentricity does not occur.
2. Rock surface to be tested should not be grinded after saw cutting.

P9- DRILLING RATE INDEX

Aim: Determination of Brittleness index, Siever's J Value and Drilling Rate Index

a) Brittleness Index

The brittleness test gives a good measure for the ability of the rock to resist crushing by repeated impacts. The test method was developed in Sweden by Matern and Hjelmer in 1943. Density for each sample is calculated by dividing the mass (measured on an electronic balance) by the volume (length and the diameter are both measured by caliper).

Test Procedure

About 500 gm crushed rock sample between 16- and 11.2-mm sieve size is taken and pounded with a 14 kg weight from a height of 25 cm for 20 times. An outline of test procedure is shown in Fig. 1. The BI value is the percentage of the material that passes the 11.2 mm sieve after 20 drops and is determined as the mean value of 3 to 4 parallel tests. The test apparatus used in the tests as show in Fig. 2. The brittleness value S_{20} is an indirect expression of the necessary amount of energy it takes to crush the rock. Sample size determined as given below:

Weight of sample (in gms) = (Density of rock sample/2.65) *500

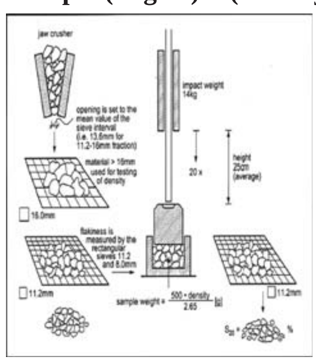

Fig.1 Outlines of principles for the brittleness test (Dahl, 2003)   Fig.2 The brittleness test apparatus

	
Initial sample of size between 11.2 - 16 mm	Crushed sample after 20 impacts of 14 kg wt.
	
Sample which passed through 11.2mm sieve	Sieving with 11.2 mm sieve size

Fig.3 - Different steps of brittleness test

The results of brittleness test are recorded as given below:

Observation Table: Test results of Swedish Brittleness Index (BI)

Rock Type	Test No.	Sample Weight (gms)	Pass weight (-11.2mm)	Brittleness Index (BI)	Average/Range BI
Quartz Feldspar Mica Gneiss	1	472	308	65	**68 (65-72)**
	2	472	316	67	
	3	472	320	68	
	4	472	342	72	

$$Specimen\ Weight\ (W_1) = \frac{500 \times Specific\ Gravity}{2.65} \qquad (1)$$

$$Brittleness\ Value\ (S_{20}) = \frac{W_2}{W_1} \times 100 \qquad (2)$$

where W_2 is the amount of crushed specimen passing 11.2 mm sieve. The Brittleness Value is the mean value of 3 - 5 parallel tests.

Inference

Consistency in the results is to be checked. Variations need to be explained considering the macro and micro features of the rock specimen. Higher brittleness, in general, is an indicator of easier Boreability/drillability. However, this needs to be read alongwith Siever's J Value and thin section analysis.

B) Siever's J Value

Introduction

The Seiver's J value expresses indirectly the surface hardness of the rock. The hole depth measured in 10^{-1} mm units after 200 revolutions by the miniature drill with thrust of 20 kg is known as Sj value. It is determined as the mean value of 3 to 4 parallel tests. The drill bit used is 8.5 mm in width with 110^{o} bevel angle. The test setup is shown in Fig.4. Specimen and bits after testing are shown in Fig.5. The combination of the Brittleness value and the Seiver's J value was developed empirically in order to give the best possible correlation between the DRI and penetration rates of drills and TBMs.

Test procedure

- Prepare tungsten carbide drill bits.
- Insert the drill bit into the chuck and tighten it. Ensure that the drill bit is properly centered in order to avoid any eccentricity.
- Examine the test surface of the specimen and mark 3-4 suitable drill spots.
- Clamp the test specimen and lower the weight carefully until the edge of the drill bit contacts with test surface. Verify that the edge is parallel with the test surface and release the lever.
- Start the rotation and run the test until the drill bit has performed 200 revolutions.

- Apply the lever and loosen the specimen from the edge of the drill bit to move to the next drill spot.

Fig.4 – Test setup for Siever's J value determination

Fig.5 – Miniature drilling bits used

The observation table for recording the results of miniature drilling are given below:

Table 1 – Test results of Siever's J Value (Sj)

Rock Type	**Test No.**	**Initial Reading (mm)**	**Final Reading (mm)**	**Penetration (mm)**	**Sj Value**	**Mean Sj Value/Range**
Quartz Feldspar Mica Schist	1	70	65	5	50	**52 (35-70)**
	2	75	68	7	70	
	3	70	64.5	5.5	55	
	4	75	71	4	40	

Inference

Higher values observed indicate higher boreability/cuttability/drillability. Variation in drillability values in similar category can be attributed mainly to mineralogical variation, their random distribution and gneissocity direction.

C) Drilling Rate Index

DRI tests on the cored samples are performed in the laboratory as per the standard procedures suggested in NTH method (NTNU, Norway). The Brittleness Index (BI) and Siever's J Values (Sj) are taken and are used to read DRI. The DRI values are determined using a nomogram developed as shown in Fig.3. The observation table for DRI is presented in Table 2.

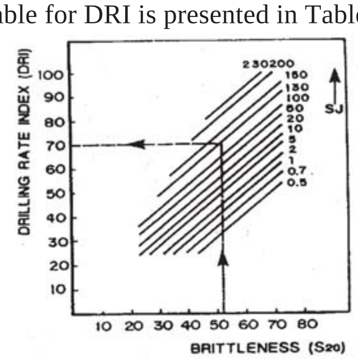

Fig.3 – Nomogram for estimation of DRI

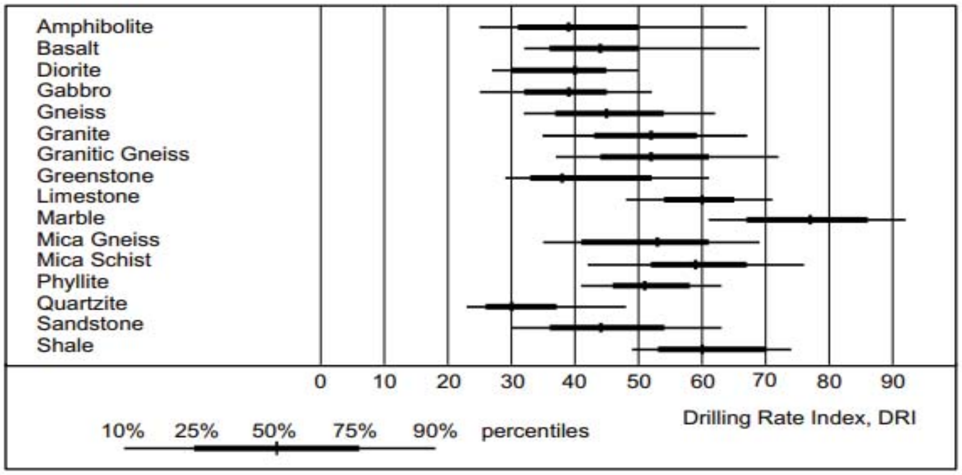

Fig.4- Typical DRI values reported (Bruland, 1998)

Table 2 – Determination of Drilling Rate Index (DRI)

Rock Type	Sample No.	Brittleness Index (BI)	Sievers J Value	DRI
Quartz Feldspar Mica Schist, Quartzo-Feldspathic Gneiss, Quartz Feldspar Biotite Schist	1,2,3,4	53	52	64

The classes into which the rock suits fall from drillability view point are shown in Table 3.

Table 3- Category intervals for drillability (Bruland A., 1998)

Category	DRI
Extremely low	<25
Very low	25-32
Low	33-42
Medium	43-57
High	58-69
Very High	70-82
Extremely High	> 82

Low DRI values correspond to difficult drilling, in general.

Reference:

Amund Breland (1998), Hard Rock Tunnel Boring, Drillability Statistics of Drillability Test Results, NTNU, Trondheim

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P10- PUNCH PENETRATION INDEX

AIM : Determination of Punch Penetration index of rock

SCOPE

The test is conducted to examine the toughness inter alia cuttability or boreability of rocks via the slope of the best-fit line on force-penetration chart. It also helps in estimation of brittleness/toughness of rock.

APPARATUS

1. Indenter: A conical shape Tungsten carbide indenter is required. The radius of tip is 3.125 mm.
2. A loading frame (capacity 500 kN) for applying force on indenter.
3. Holder- Sample holding cylinder. (Fig.1)
4. Load cell up to 500 kN capacity.
5. LVDT: Linear variable displacement transformer to measure penetration.
6. Data Acquisition System

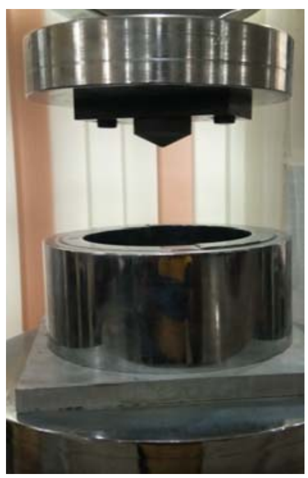


Fig.1 – Punch penetration cell with indenter

INTRODUCTION

The punch penetration test, originally intended to provide a direct method for estimating the normal load on disc cutters was developed in late 1960s to provide direct laboratory method to investigate rock behavior under the indenter (Handewith, 1970). Yagiz (2002) used the test results as a brittleness index, one of the input parameters for Modified CSM model, to estimate the TBM penetration. Yagiz (2009) also stated that the punch penetration test that could be directly used for measuring rock brittleness has a great potential to estimate the penetration rate of tunneling machines.

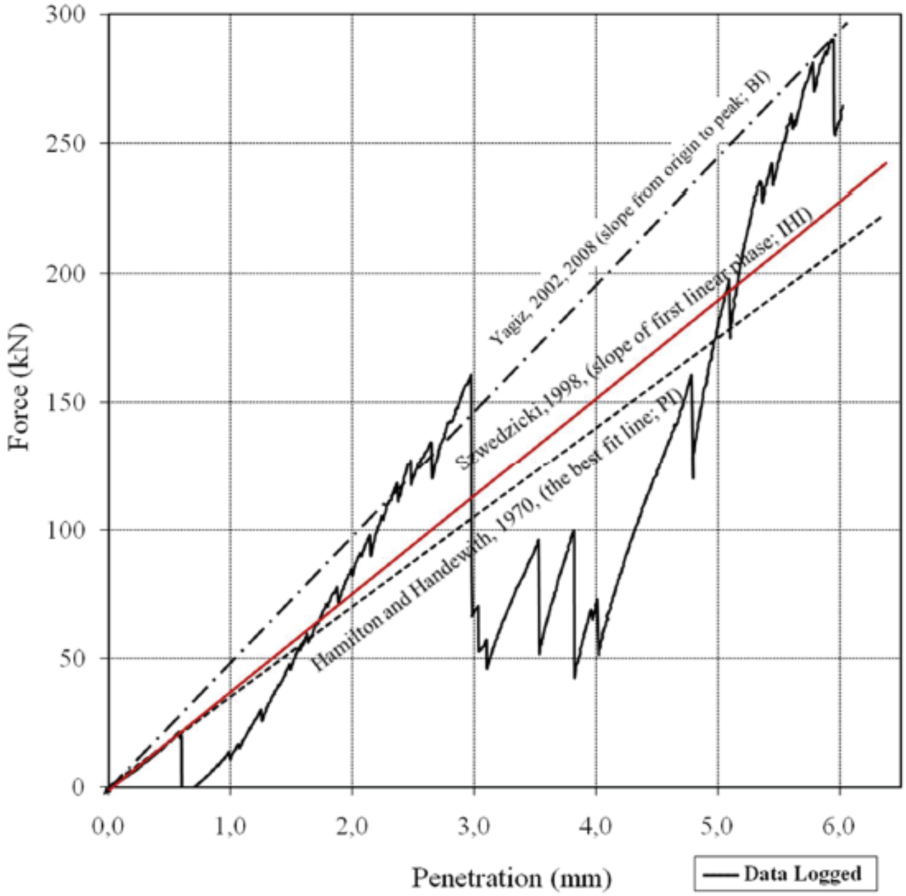


Fig. 1 Various Expressions from the Fluctuated Force-Penetration Graph of the Test

The slope of the best fit line (kN/mm or lb/in) was named as penetration index (PI) employed for predicting the expected cutter force and corresponding penetration for mechanical excavation (Hamilton and Handewith, 1971).

In the first phase, there is elastic deformation and very fine crushing of rock surface. In the second phase, there is crushing of rock fabric as in the third phase chips of rock are formed. Elastic deformation and very fine crushing is represented by a linear relation. Crushing in represented by "steps" on the profile and chipping of the rock fragments is represented by peaks (Fig. 2). Consequently, the brittleness index (BI) in kN/mm was computed as using the slope of whole phase of force-penetration profile, obtained by drawing line from origin of the chart to the maximum applied force that rock absorbs till the test ended (Yagiz, 2009). As result, the brittleness may be computed as follow;

$$BI = \frac{F_{max}}{p} \quad (1)$$

Where, F_{max} is maximum applied force on a sample in kN, and P is corresponding penetration in mm.

In this graph, the higher brittle rock shows fluctuated force-penetration profile due to the large force drops with large chips; though, moderate brittle rock indicates minor force drop with small chips. As the rock is ductile or low brittle, then there is no force dropping and chipping on the rock but crashing. Further, based on obtained values and fluctuated force-penetration profile from the punch penetration test, the rock brittleness could be classified (Table 1).

Table 1 Rock Brittleness Classification Obtained from The Punch Penetration Test (Yagiz, 2009)

Brittleness index (kN/mm)	Brittleness class
≥ 40	Very high brittle
35- 39	High brittle
30 -34	Medium brittle
25-29	Moderate brittle
20 - 24	Low brittle
≤19	No-brittle (ductile)

PROCEDURE

1. Install the setup using load cell, LVDT and data acquisition system.
2. Prepare the cylindrical shape rock sample of height 80 mm with saw cut ends.
3. Place the holder on the load cell and put the rock sample inside holder at the center.
4. Adjust the platen fitted with indenter to maintain contact with rock surface.
5. Set the loading rate at ____kn/min
6. Start the machine by pressing green button and simultaneously start the data acquisition.
7. Record the force vs. penetration graph until the rock fails and stop the machine by pressing red button.
8. Repeat the above experiment from step 2 with another 3 – 4 samples and record the results.

OBSERVATIONS

SN.	Rock Type/Sample no.	F_{max} (kN)	P (mm)	BI (kN/mm)
1.				
2.				
3.				
4.				

CALCULATION

$$BI = \frac{F_{max}}{p} \text{ kN/mm}$$

Inference: Draw suitable inferences with regards to the rock classification with respect to brittleness as explained.

***

P11- LINEAR CUTTING RIG EXPERIMENTS

Aim: Determination of cutting forces on disc cutter using liner cutting rig and arrive at specific energy for cutting a given rock

Equipment: Linear cutting rig, Triaxial force transducers, thrust and linear hydraulic rams, instrumentation, data acquisition system etc.

Mechanism of disc cutting:

Disc cutters are used in TBM for cutting hard rock in tunneling and mining. They cut concentric groves on the tunnel face and the breakage is effected predominantly by tensile fracturing and chipping. The mechanism of cutting is shown in Fig.1.

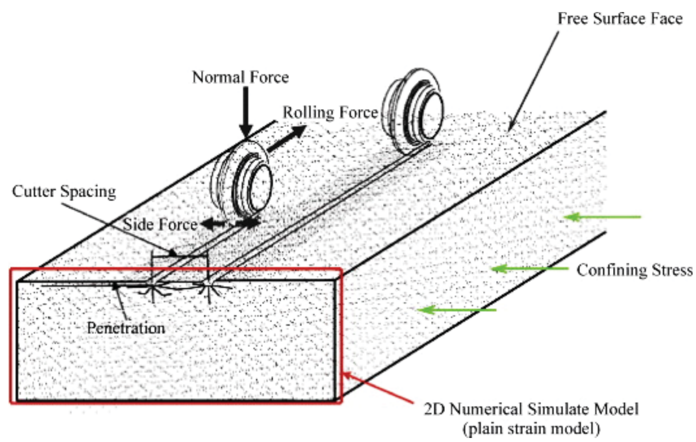


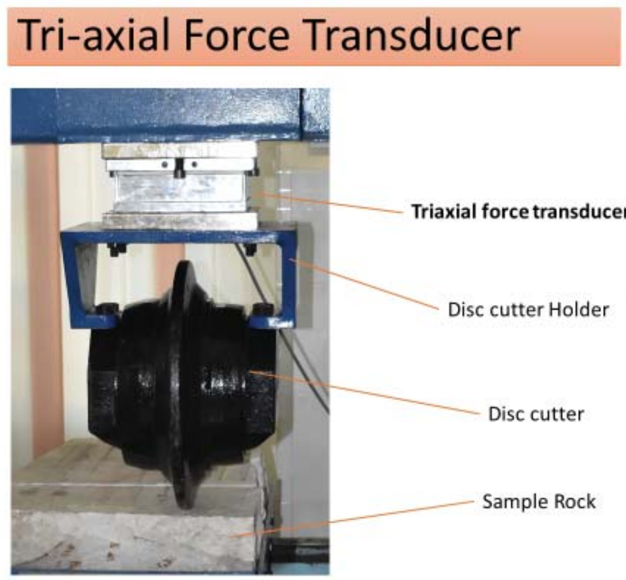
Fig.1 – Mechanism of disc cutting using linear cutting rig

Linear Cutting Rig is designed and developed by Rock Excavation Laboratory, department of Mining Engineering, IIT(ISM) Dhanbad for estimating the specific energy required for generating tensile fracture for a particular type of rock. This is also useful to arrive at the optimum spacing to depth ratio that yields lowest specific energy.

Sensor installed on the machine:

- **Draw wire sensors**- Two draw wire sensors (HBM) are installed on the LCR setup. One vertical sensor for measuring the penetration of the disc cutter in to the rock. One horizontal setup which will give the horizontal movement and show the length of cut on the rock surface. Both the sensors will give the readings in mm.
- **Tri-axial force transducer:** This is a strain gage-based force transducer for measuring force in X,Y,Z axis . In X-axis it will measure the rolling force, in Y-axis it will measure the side force and in Z-axis it will measure the vertical thrust acting on the rock through the disc cutter.

This is a custom-made sensor which can measure load up to 40 tons in vertically(Z-axis), 20 tons in X direction. It is clamped below the hydraulic press by nut and bolt arrangement and below this the disc cutter is fitted with the help of a fixture. The specifications of triaxial transducer and disc cutter arrangement is shown in Figure 2. Disc can be replaced by a pick and in such a case the hydraulic cylinder for horizontal movement needs to be replaced for imparting a higher linear velocity



Tri-axial Force Transducer	
Sensor	Strain-gage based
Vertical load range (Fz)	40 tons
Longitudinal load range (Fx)	20 tons
Transverse load range (Fy)	20 tons
Resolution	10 N
Accuracy	±0.1%

- Calculation of Specific Energy (SE)
- Measurement of forces
  - Rolling force
  - Side force
  - Reaction force

Fig.2 – Triaxial force transducer and the specifications of the testing

Procedure

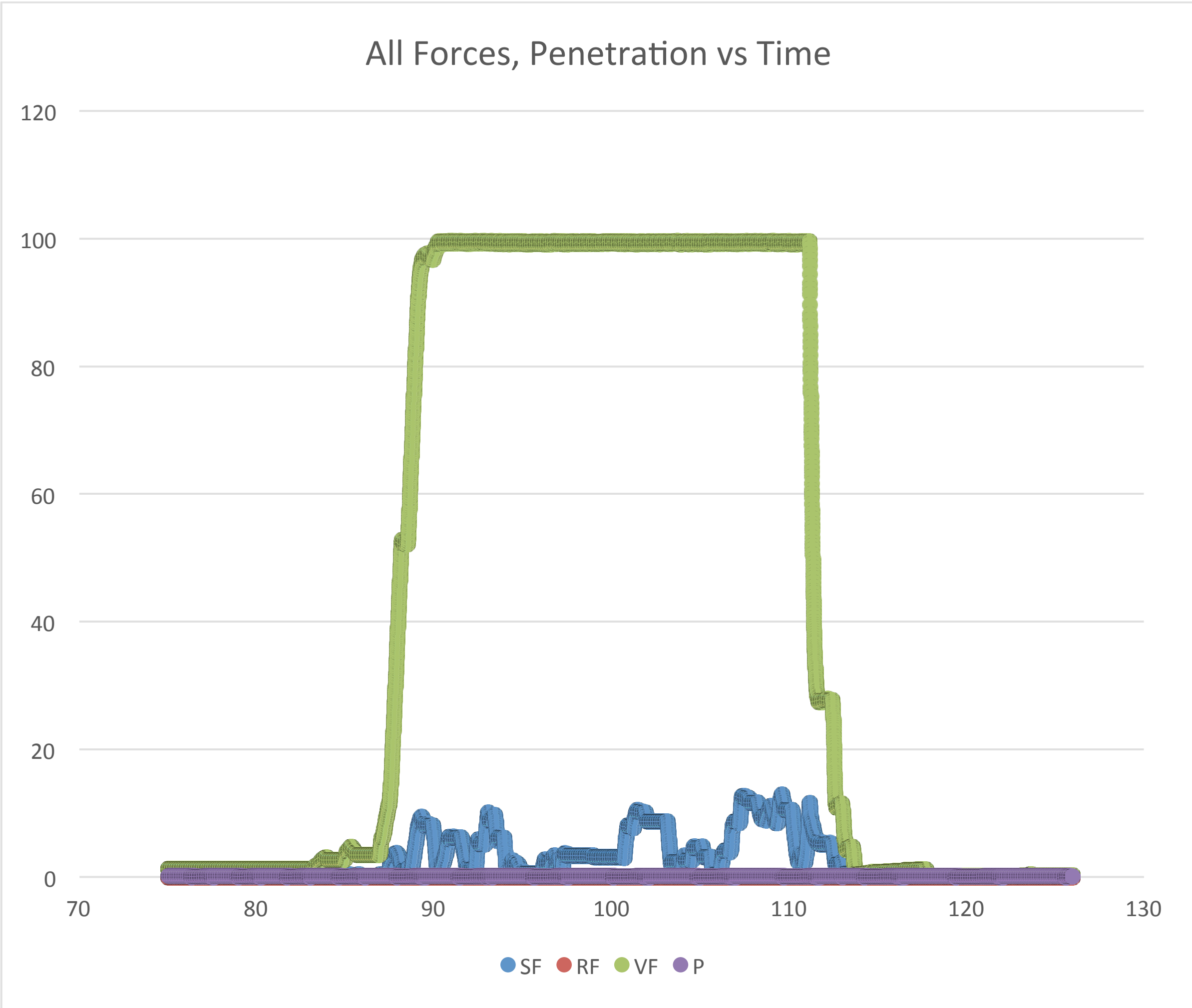
In TBM excavation disc cutter used is the primary tool for excavation. It is also known as roller cutters, which rolls on a specific radius on the cutterhead and generates crack, which in turn generate chips. Therefore, in this experiment a 17-inch disc cutter is used for eliminating any scaling factors. This 17-inch disc cutter is of Terratec make having edge width of 19.5 mm. It is a constant cross section type disc cutter. The metal ring is of H13 tool steel . The metal sample tray is used to hold the rock sample in its position. The sample tray has a width of 520 mm and a length of 820 mm.

After clamping the rock sample on the sample tray lines with the help of chalk or marker are drawn on the rock surface with the desired spacing (Ex: 25mm, 50mm and 75 mm). After the lines are drawn the cutter is brought down and touched on the rock surface. The disc cutter is brought to starting position where edge effect is zero. Then the required thrust force is applied on the disc cutter (Ex:2 ton, 4 to,6 ton, 8 ton). After applying thrust the penetration is recorded and visualized using CATMAN software through HBM data acquisition system which is installed on the computer connected to LCR.

The applied thrust is recorded and visualized using the display connected to tri-axial sensor. After applying required thrust the disc cutter starts its movement from one point to another point on the pre-drawn line with the applied thrust. The tray moves from one end of the machine to the other end generating a cut on the rock. Sensors are fitted on both the ends of the machine to limit the movement of the trolley within the safe operating distance.

Experiments were conducted on Manikaran quartzite which is highly brittle and having a very high strength. The forces and penetration observed are depicted in Fig.3.

Data Analysis:



- As we can see from the graph the vertical force is high but the penetration is very low as the rock strength is very high.
- Vertical force acting on the rock was 99.674 kN.
- Side force acting on the rock 12.873 kN.
- Rolling force and penetration are very low.

Typical, Spacing is 25 mm. Penetration is 2 mm. Therefore, s/d ratio is 12.5.

Specific energy is calculated from the side forces which cause chipping.

The experiments need to be carried out with varied spacing to depth ration and the specific energy are computed. A plot relating s/d ratio and specific energy is shown in Fig.4.

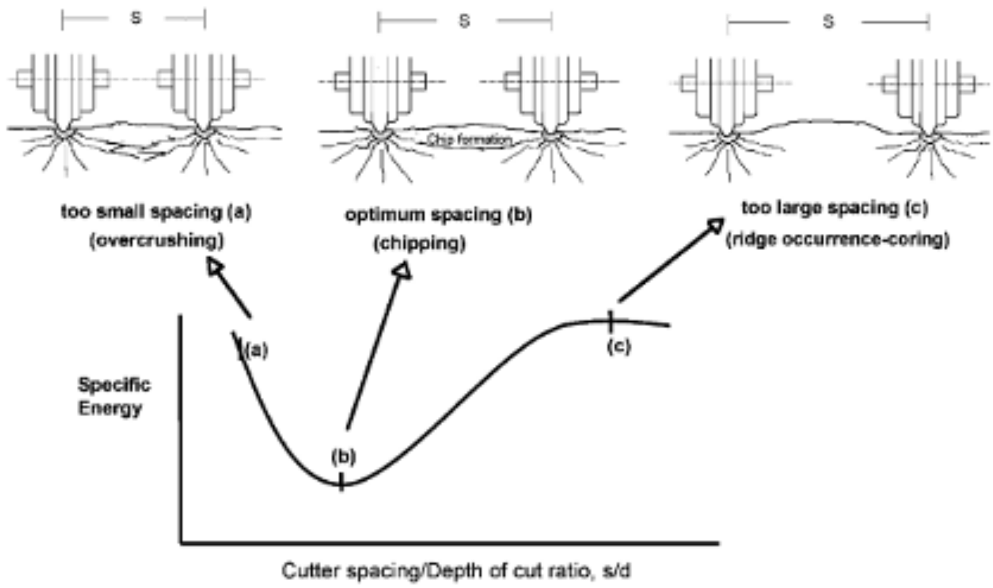


Fig.4 – Effect of s/d ratio on specific energy

The specific energy is calculated by the following expression:

$$SE = \frac{FR}{Q}$$

where SE is the specific energy in MJ / m3, FR is the force acting on the cutting disc in kN and Q is the volume per unit length of cut in m3 /m. It is also possible to determine the instantaneous speed of cutting as:

$$ICR = k \cdot \frac{P}{SE_{opt}}$$

where ICR is the instantaneous speed of the rock cutting in place in m3 / h, P is the power in kW, SEopt is the optimum specific energy obtained from Rock Cutting Test in kWh / m3 and k is a constant depending on the efficiency of the system , expressed as a ratio of the energy transferred from the excavating head to the surface of the front and between 0.85 and 0.9.

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P12- BIT WEAR INDEX, ABRASION VALUE STEEL AND CUTTER LIFE INDEX

AIM

Determination of Bit wear index, abrasion values steel and computation of cutter life index (CLI)

SCOPE

These parameters are a measure for the time dependent abrasion on tungsten carbide/ cutter steel in the presence of crushed rock powder.

INTRODUCTION

The test method was developed at the Department of Geology at NTNU in the years 1958 – 61 (Reidar Lien, Rolf Selmer-Olsen). It estimates the abrasivity of rock. This test is used to calculate the Bit wear index (BWI), Abrasion Value Steel and Cutter life index (CLI) for drilling bits and disc cutters respectively. Thus, it helps in estimating the cutting life of tools during excavation. BWI is calculated by Abrasion value (AV) test which requires testing with tungsten carbide bit for 5 min at a speed of 20 rpm. CLI is estimated using Abrasion value steel (AVs) test which is calculated by testing of cutter ring steel bits for 1 min at 20 rpm speed.

APPARATUS

1. Test Specimen: Rock samples to be tested for their abrasivity level are crushed into powder with particle size less than 1 mm. Prepare 500 g of rock powder.
2. Vibrating feeder: A feeder is required to pour the test specimen on a rotating disc at a rate of 80 g/min.
3. Tool Bits: For AV test, Tungsten carbide (hardness: $HV_{40} \approx 1300$) made bits with radius of curvature as 15 mm. Tool steel (disc cutter steel) made bits with radius of curvature as 15 mm for AVs test.
4. Static load: A 10 kg load is required to act as thrust on bits.
5. Vacuum Cleaner: Rock powder after passing beneath the bits is sucked by the pump.
6. Weighing machine: A high precision balance of readability 0.001 g is required to measure the weight of bits.

Fig.1 – Bit wear index (BWI) or Abrasion Value Steel (AVS) apparatus

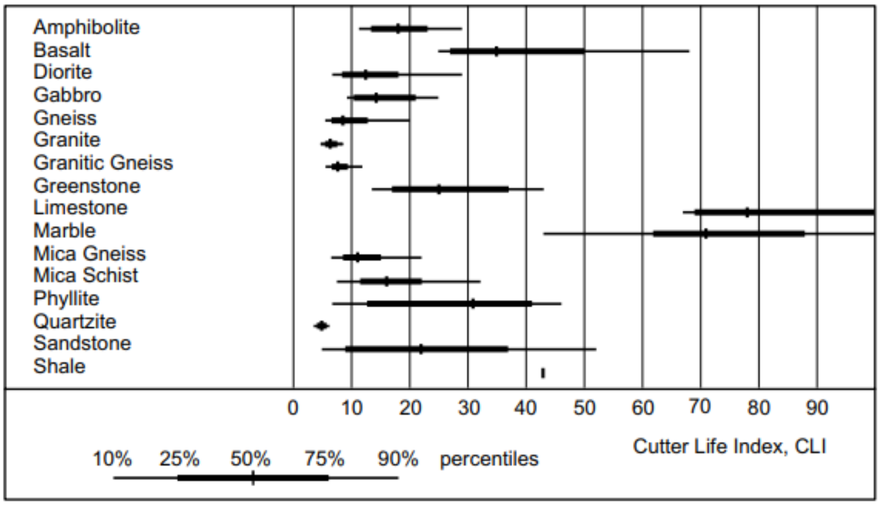

Fig.2 -CLI for some rocks (Bruland 1998)

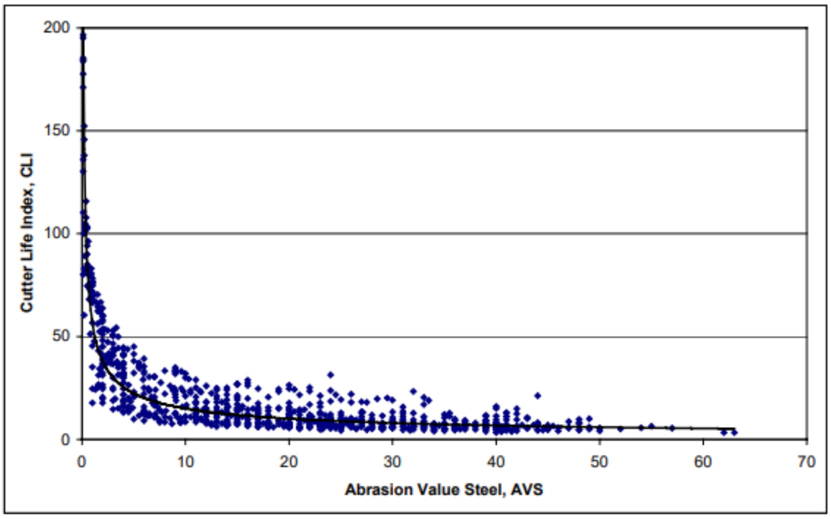

Fig.3 -Relation between AVS and CLI for some rocks (Bruland 1998)

PROCEDURE

1. Crush the rock sample into powder form with particle size less than 1 mm and put in the vibrating feeder cone.
2. Grind the tool bit to obtain the radius of curvature as 15 mm and measure its initial weight W_1 mg.
3. Attach the tool bit below the static load in the tool holder and gently place the loaded setup on the rotation steel disc. Ensure that alignment is perfectly vertical.
4. In the control panel box of machine, adjust the feeder rate to 80 g/min by trial and error. Collect the powder falling from the feeder for 1 min and measure its weight, it must be 80 g. Adjust the vibration rate by rotating the circular black knob.
5. Set the time limit for rotation of disc in the control panel as required by the tests. Also set the speed to 20 rpm.
6. Lift the loaded tool holder setup slightly above the disc and press the Start button and turn on the vibrating feeder.

7. As soon as a complete ring of rock powder on disc is formed, keep the tool setup on the disc gently. Simultaneously switch on the vacuum cleaner.
8. After the rotation is complete, clean the tool bit and weigh it (W_2 mg).
9. Repeat the procedure from step 2 for another 4 test values.

OBSERVATIONS

SN.	Rock Type	W_1 (mg)	W_2 (mg)	Loss(mg)	AV/AVs
1	Quartzite	45.761	45.746	0.015	15
2	Metabasics	45.348	45.322	0.026	26

CALCULATION

The Abrasion Value is the weight loss in milligrams of the test bit after 100 revolutions of the steel disc. 100 revolutions equal 5 minutes test time. Abrasion Value Steel is the weight loss after 20 revolutions of steel disc.

AV or AVs = W_1-W_2

Cutter Life Index

This is determined from the following equation suggested by NTNU (A.Bruland, 1998):

CLI= 13.84 (Sj/AVS)$^{0.3847}$

Values obtained for Sj in previous experiments can be taken for CLI computation. Effect of CLI on cutter ring life is shown in Fig.4.

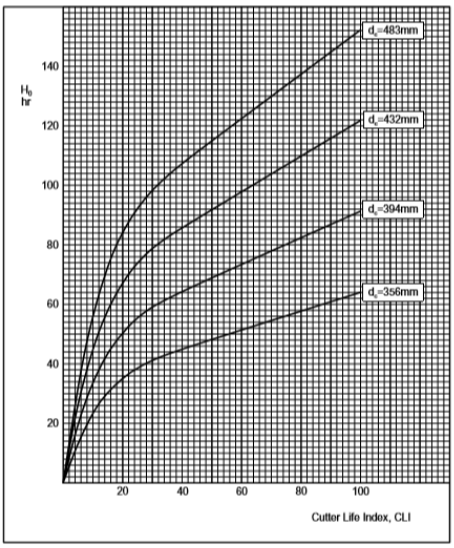

Fig.4 – Effect of CLI on cutter ring life in cutters of different diameter (A. Bruland, 1998)

Inference: Based on the CLI values the cutter life in hours can be determined.

Reference:

Amund Breland (1998), Hard Rock Tunnel Boring, Drillability Statistics of Drillability Test Results, NTNU, Trondheim and Advance Rate and Cutter Wear manuals.